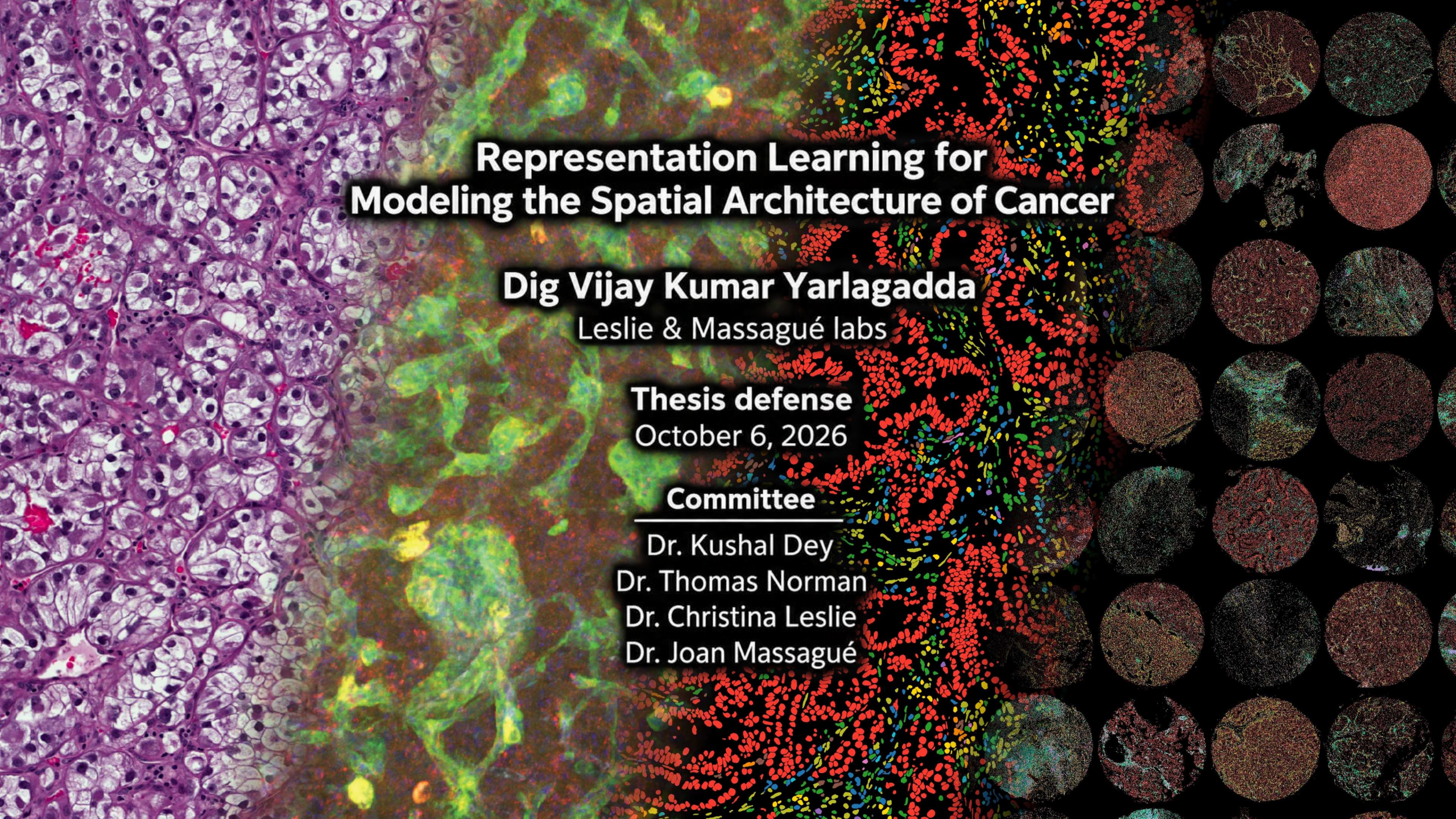


Representation Learning for Modeling the Spatial Architecture of Cancer

Dig Vijay Kumar Yarlagadda

Leslie & Massagué labs

Thesis defense

October 6, 2026

Committee

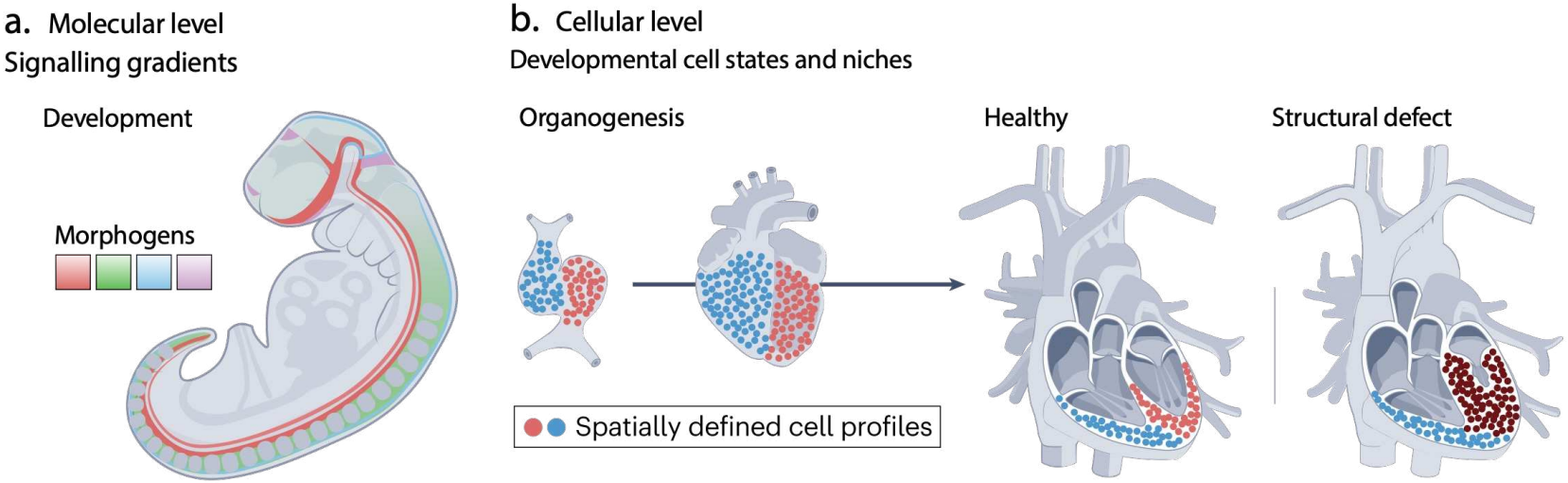
Dr. Kushal Dey

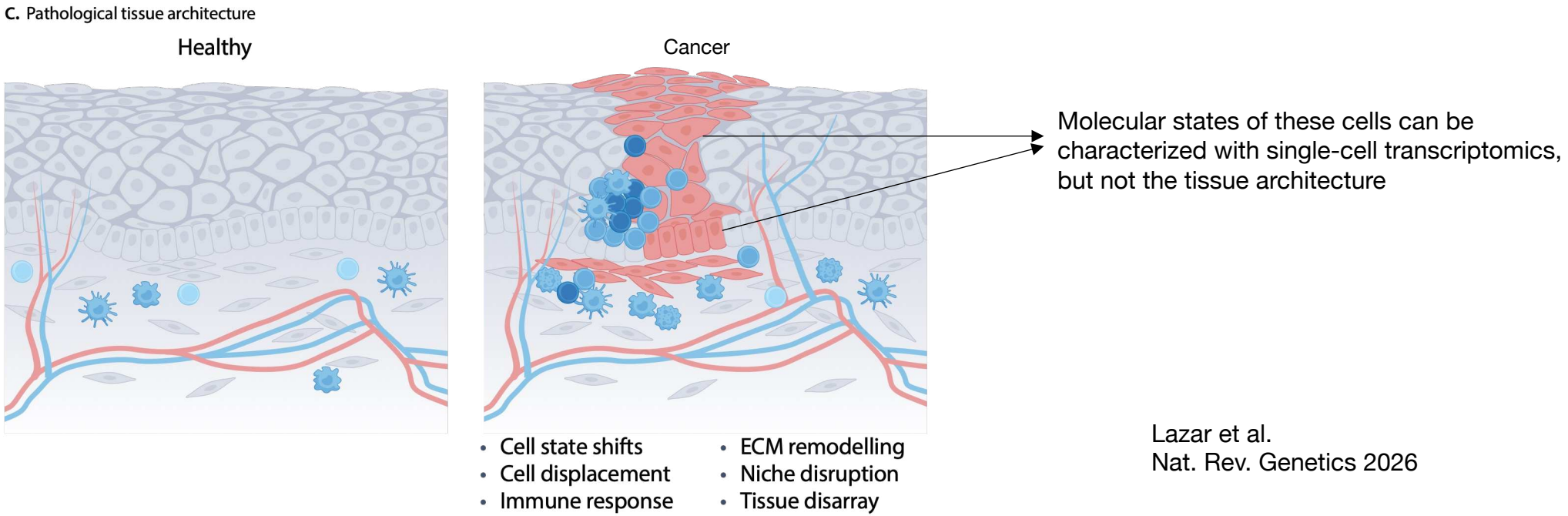
Dr. Thomas Norman

Dr. Christina Leslie

Dr. Joan Massagué

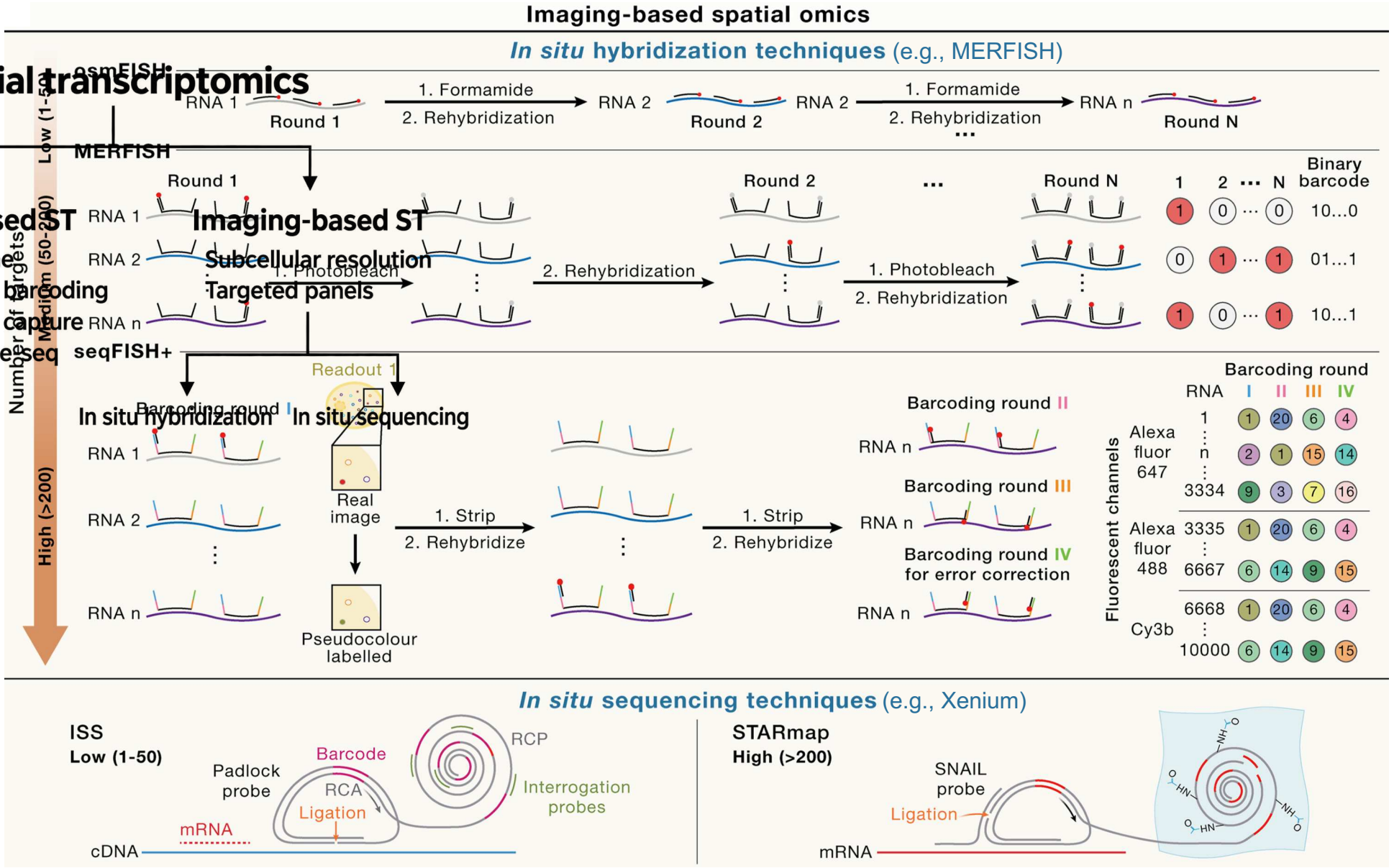
Spatial context shapes cellular identity & behavior





Lazar et al.
Nat. Rev. Genetics 2026

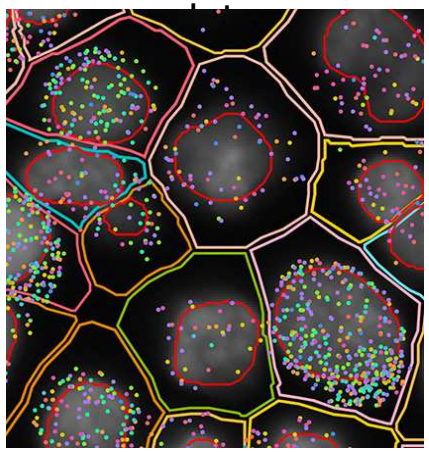
Spatial transcriptomics measures molecular state within tissue architecture



Liu et al. *Cell* 2024

imST introduces challenges in cell segmentation, sparsity and panel biases

High-throughput, high-resolution



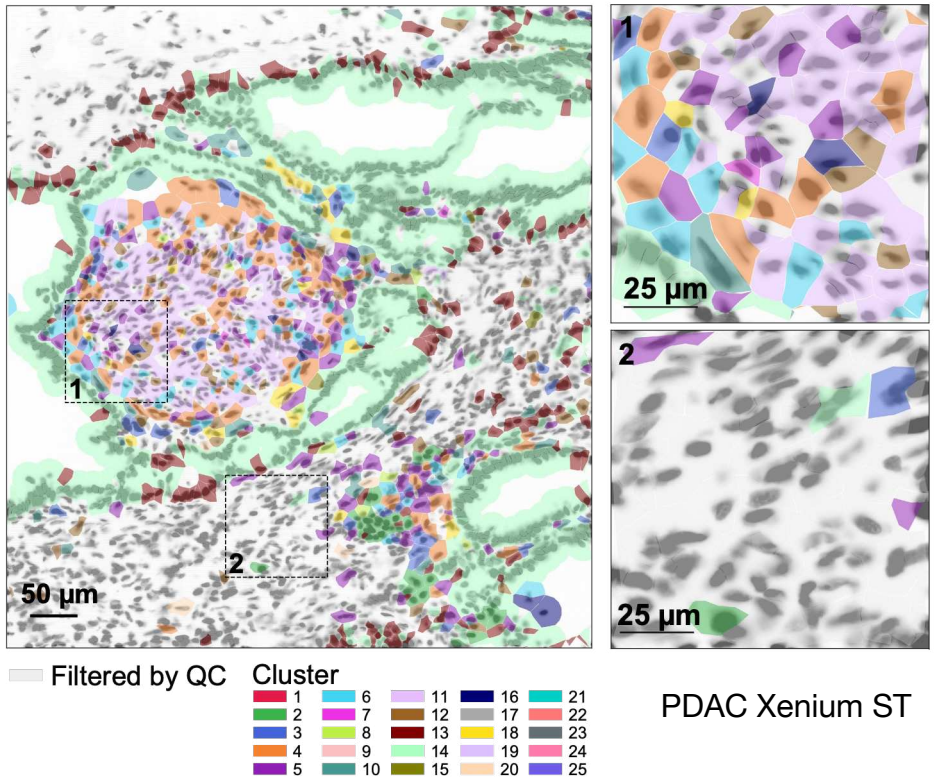
Xenium ST, RCC-BrM dataset

Challenges of imST data

- Segmentation artifacts
- Targeted panels
- 3D structure
- Transcript diffusion
- Probe sensitivity

How can we robustly infer cellular states of cancer cells & their spatial context from imST data?

Conventional preprocessing can remove cell populations of interest



PDAC Xenium ST

Outline

1. Nicheverse: Modeling Spatial Architecture using Discrete Representation Learning

- **Yarlagadda D**, Massagué J, Leslie C. *Discrete representation learning for modeling imaging based spatial transcriptomics data. Proceedings of the ICCV Workshop on Bioimage Computing* 3848–3857. 2023.
- **Yarlagadda D**, Leslie C. *CellSplat: End to End 3D Gaussian Splatting for Cellular Decomposition of Spatial Transcriptomics. CVPR Workshop on End-to-End 3D Learning*. 2026.
- **Yarlagadda D**, Massagué J, Leslie C. *Nicheverse: discrete representation learning framework for modeling spatial transcriptomics data*. Manuscript in preparation.

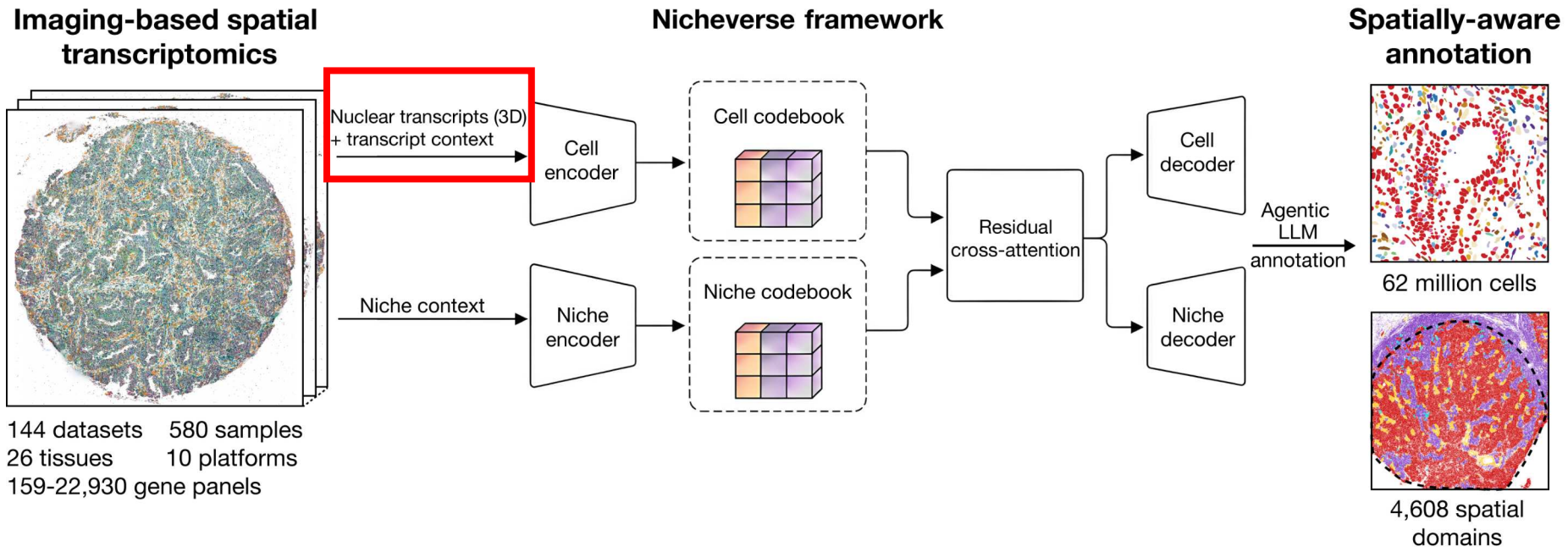
2. Developmental Reversion Underlies Resistance to Immune Checkpoint Blockade in Kidney Cancer

- **Yarlagadda D**, Wang Z, Jiang H, Vuong L, López Sanmiguel A, Yang C, Kotecha R, Chen Y, Hakimi A, Leslie C, Massagué J. *Developmental reversion underlies resistance to immune checkpoint blockade in kidney cancer. bioRxiv*. 2026.

3. Invasion Status Stratifies the Composition of the Human Pancreatic Cancer Perineural Niche

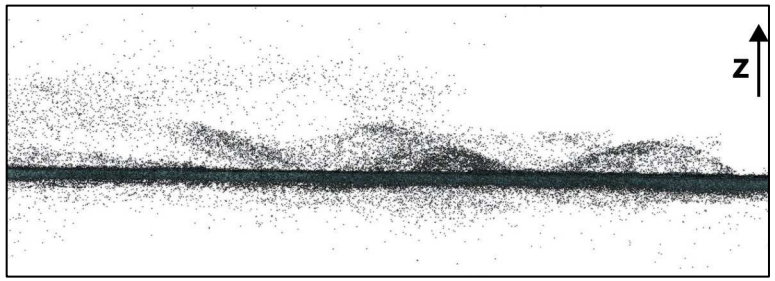
- Hussain Z*, **Yarlagadda D***, Li Q, Tezcan N, Stupakov P, Sadatrezaei G, Wong R, Massagué J, Tarcan Z, Basturk O, Deborde S, Leslie C, Sherman M. *Invasion status stratifies the composition of the human pancreatic cancer perineural niche*. *bioRxiv*. 2026.

Nicheverse: discrete representation learning for modeling imST

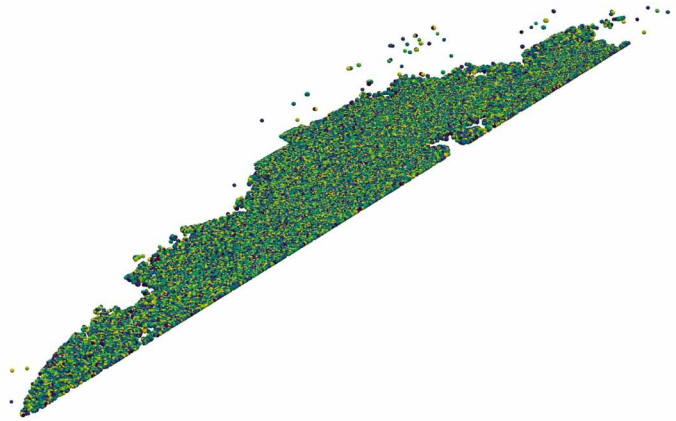


- No subjective preprocessing steps (e.g., count-based filtering, doublet removal etc.,)

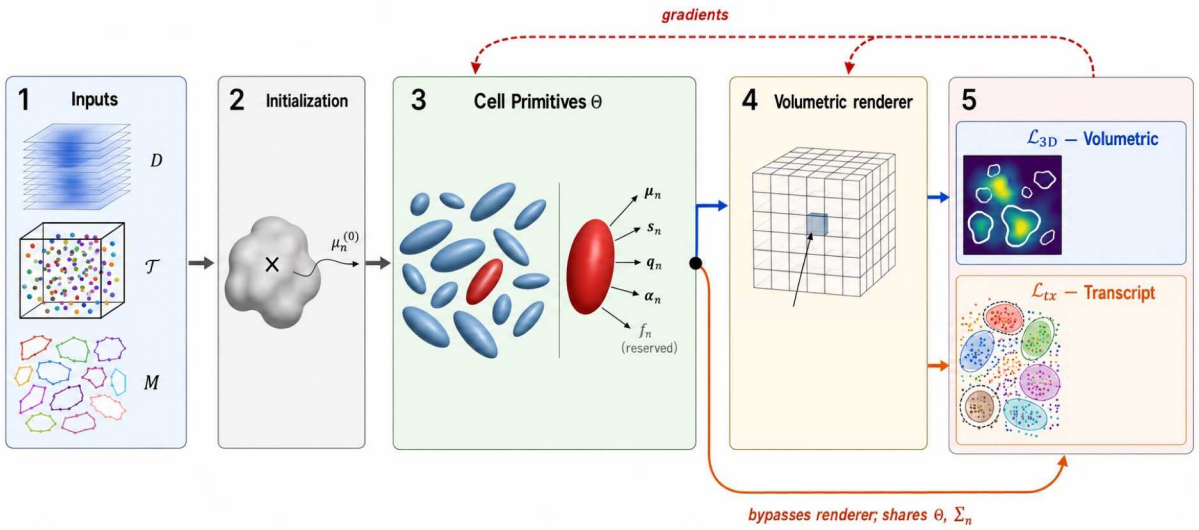
CellSplat models the three-dimensional structure of imST



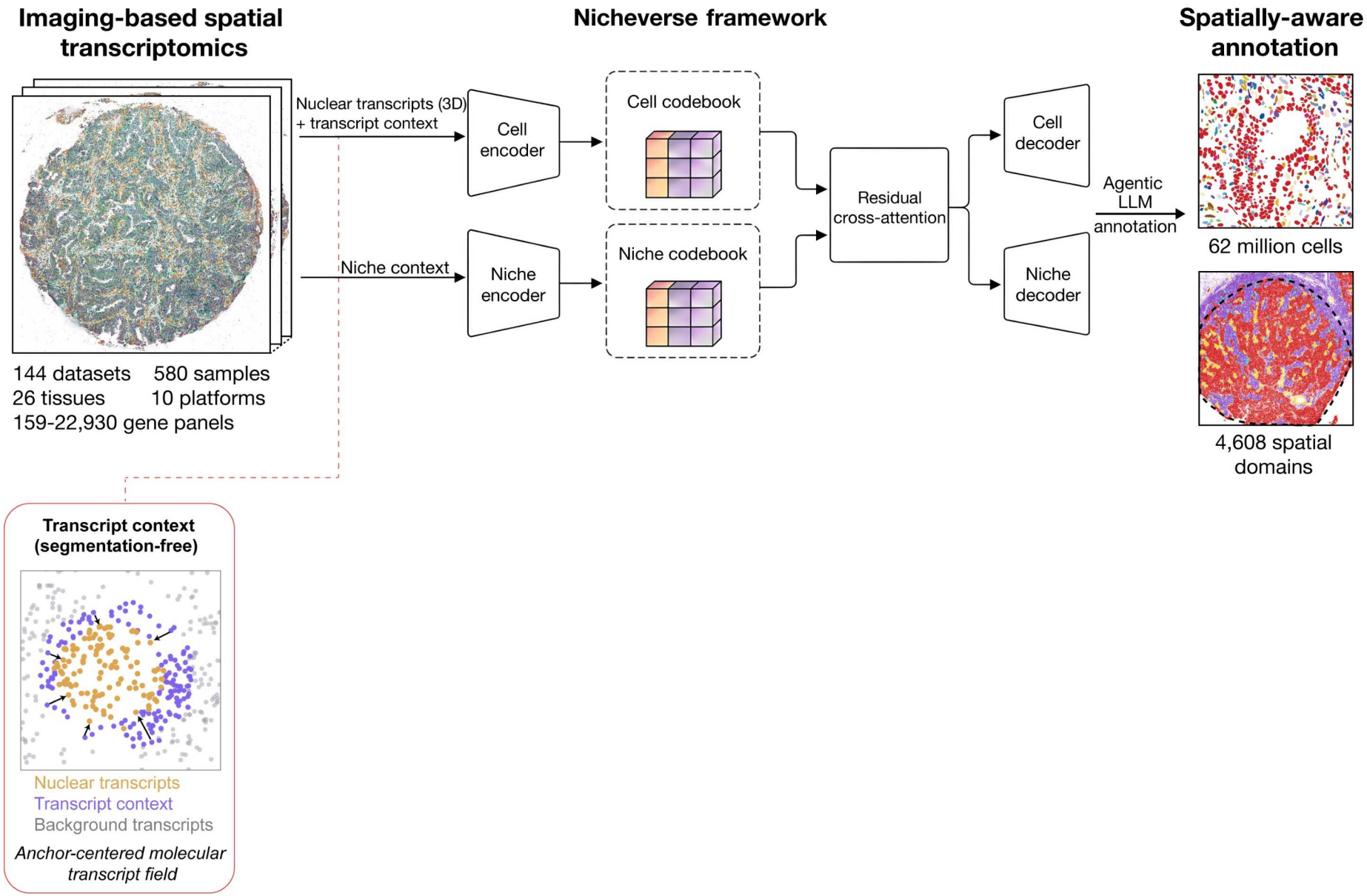
Collapsing 3D measurements into 2D can lead to cells with chimeric expression profiles



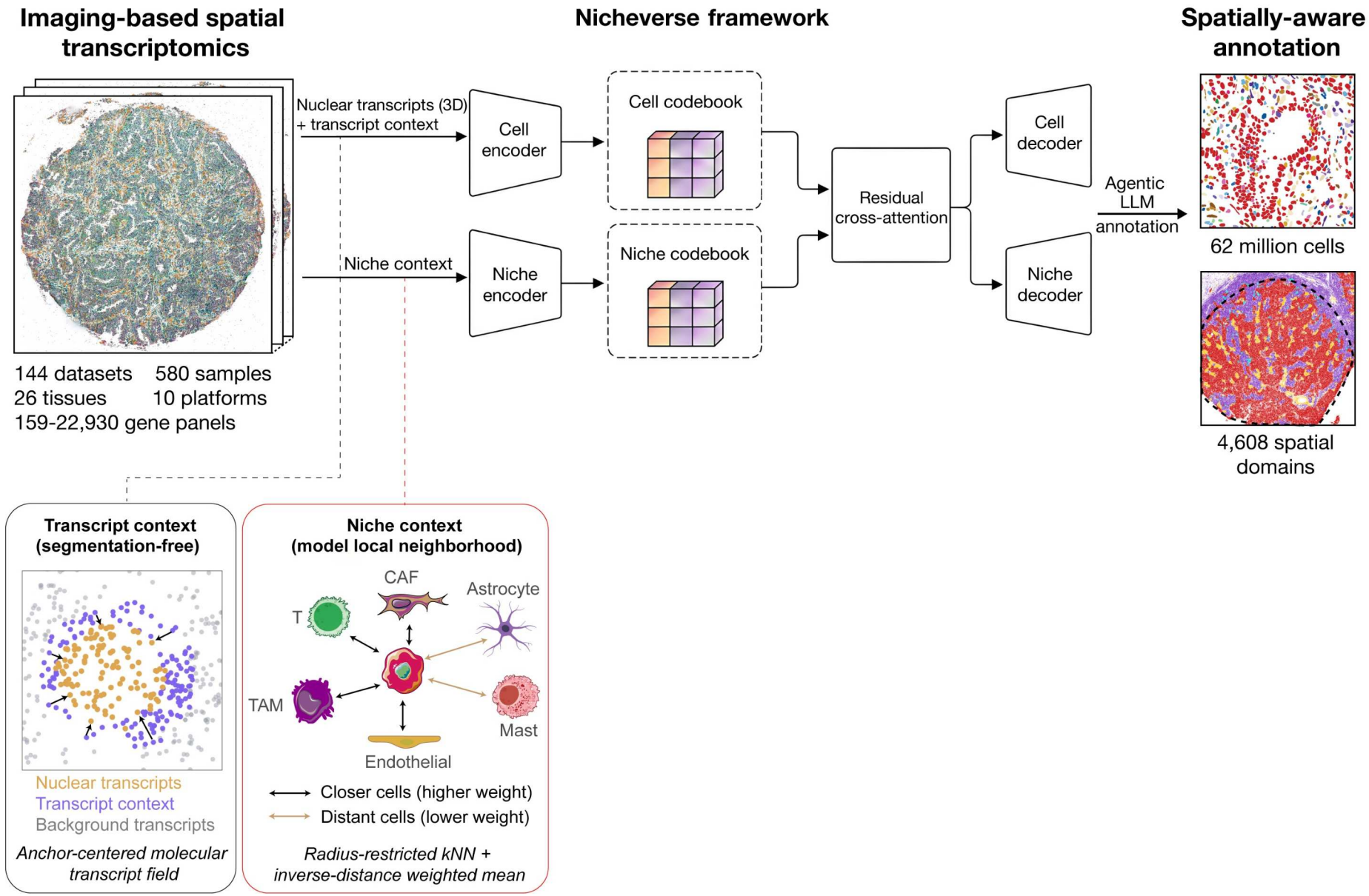
CellSplat



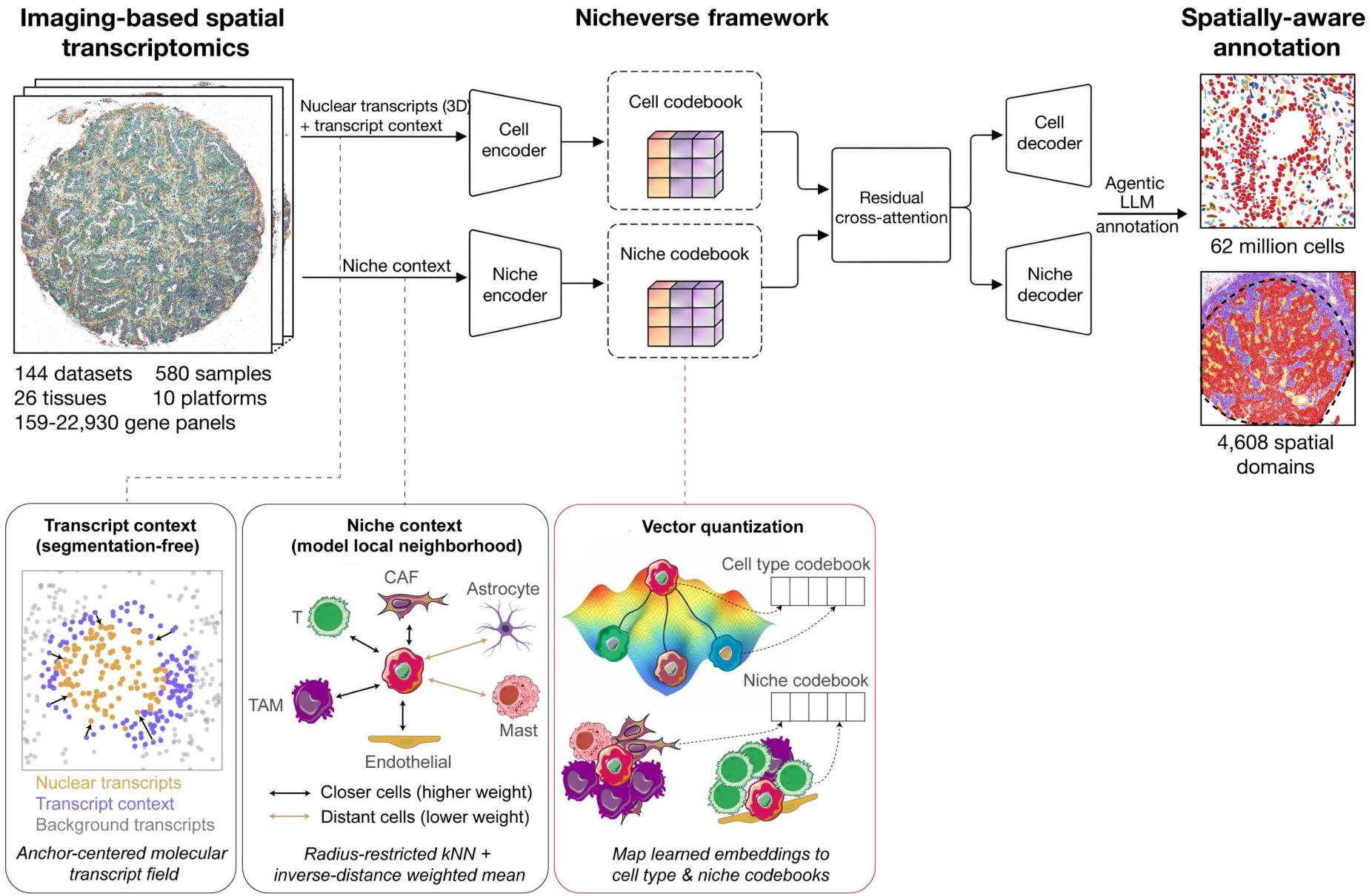
Nicheverse: discrete representation learning for modeling imST



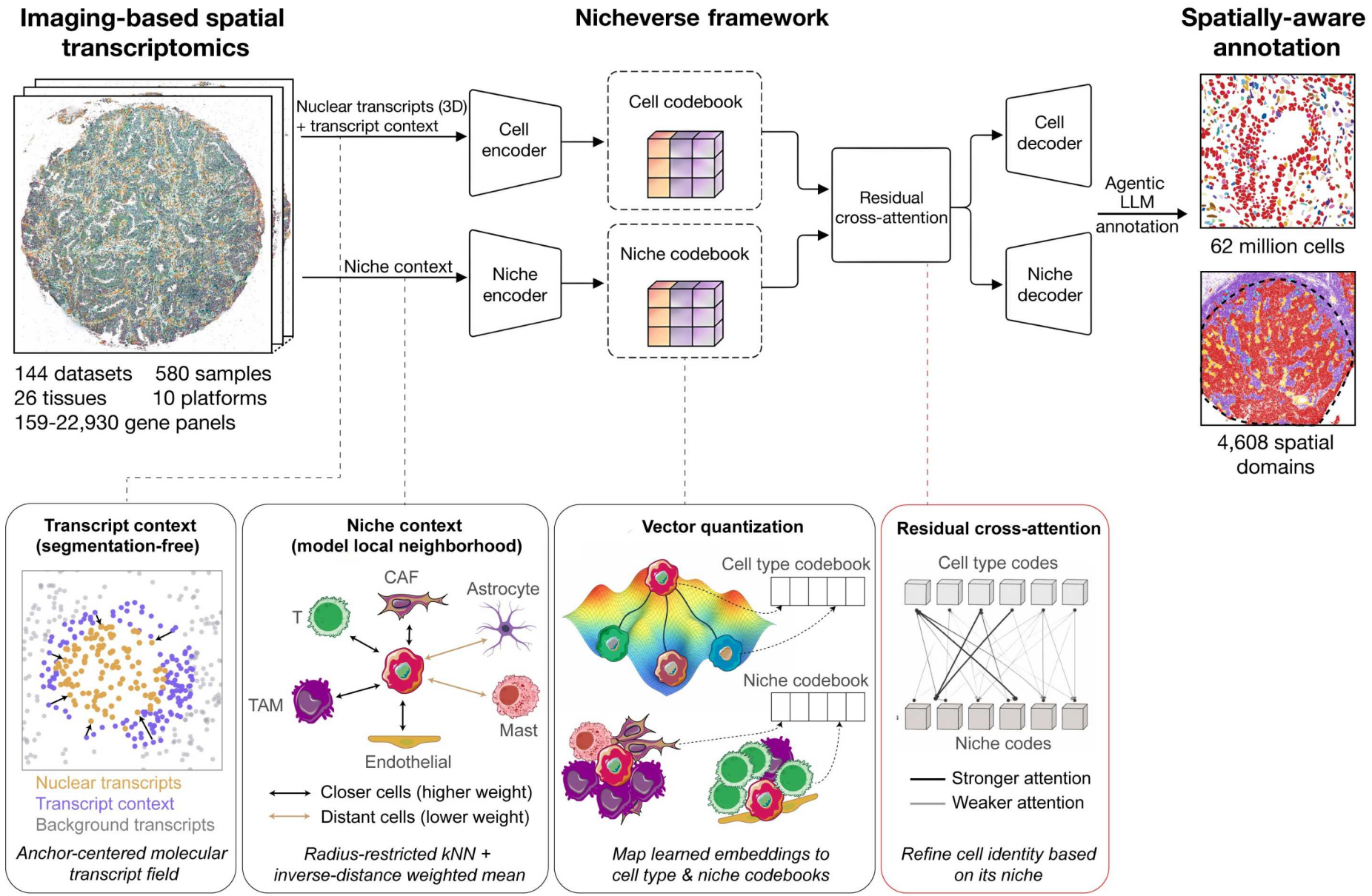
Nicheverse: discrete representation learning for modeling imST



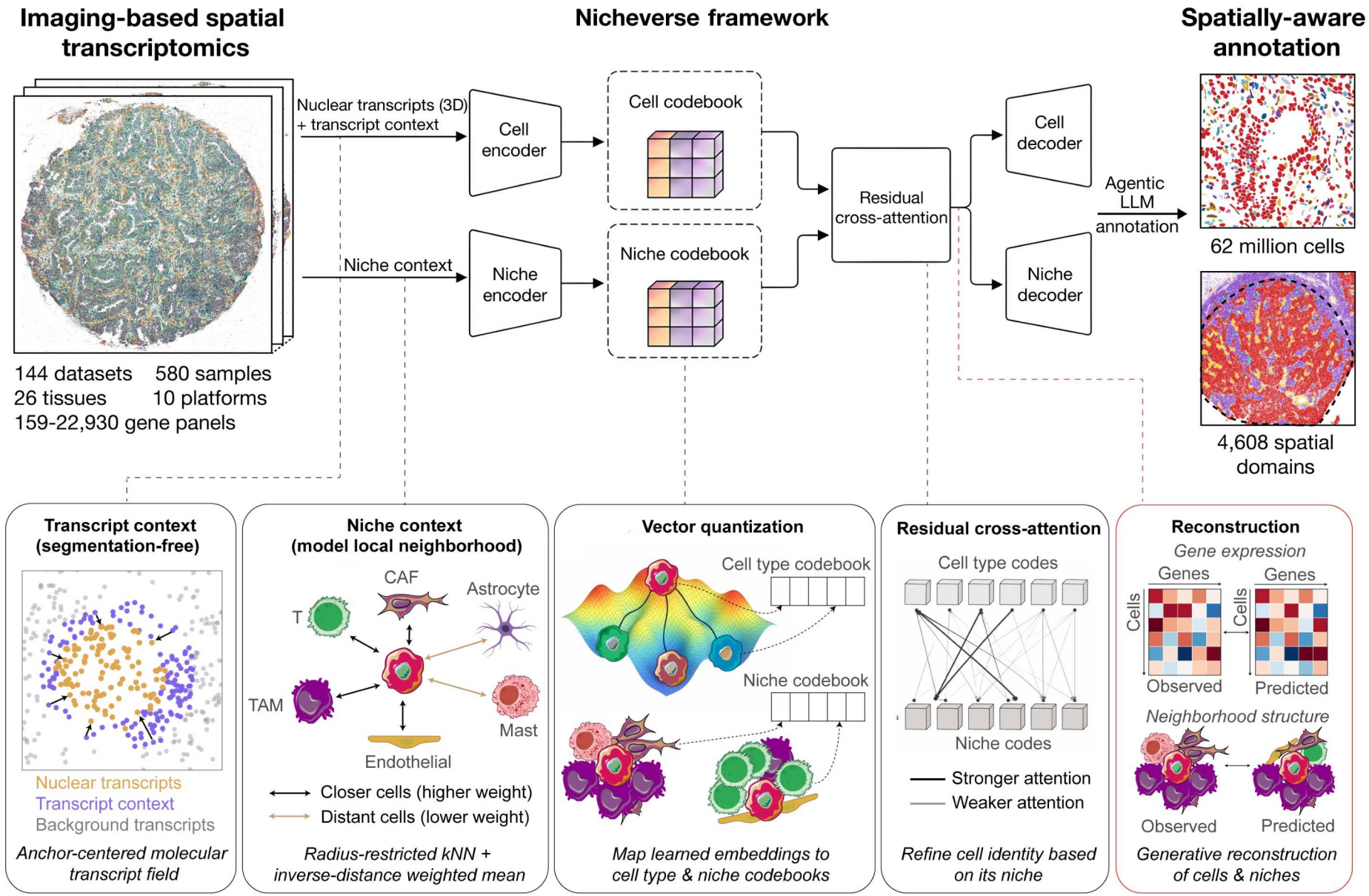
Nicheverse: discrete representation learning for modeling imST



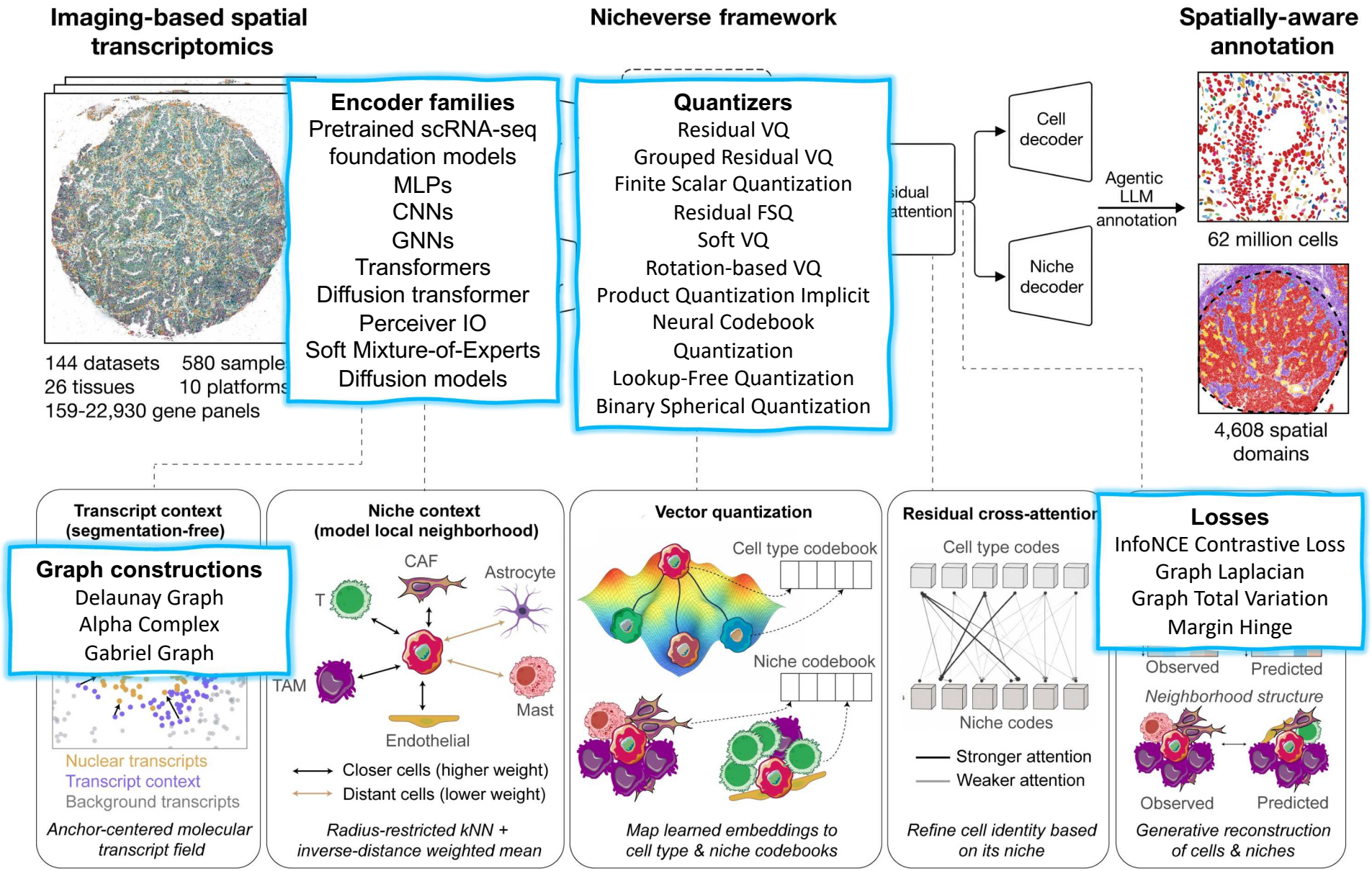
Nicheverse: discrete representation learning for modeling imST



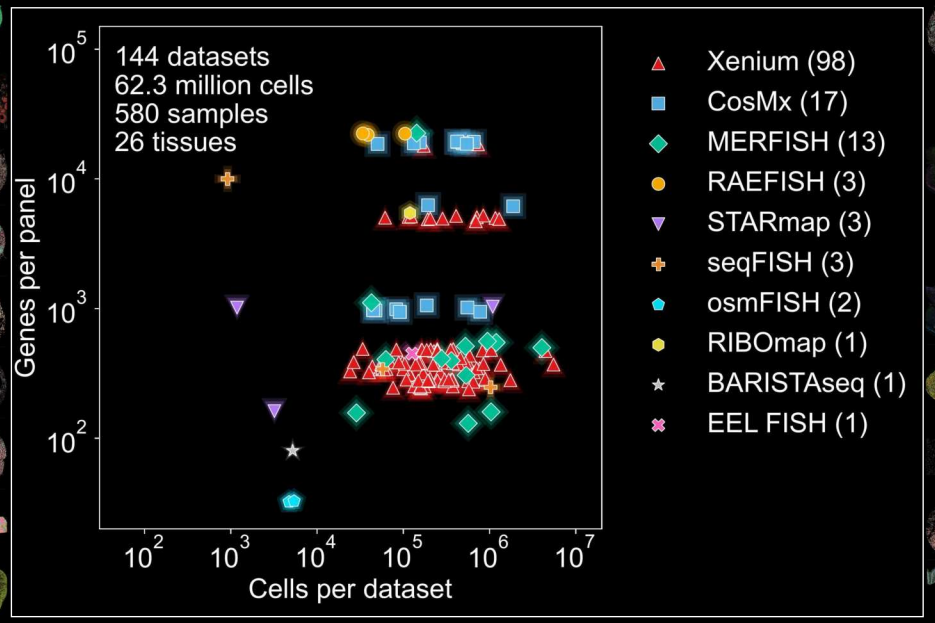
Nicheverse: discrete representation learning for modeling imST



Nicheverse is modular across encoders, quantizers & objectives

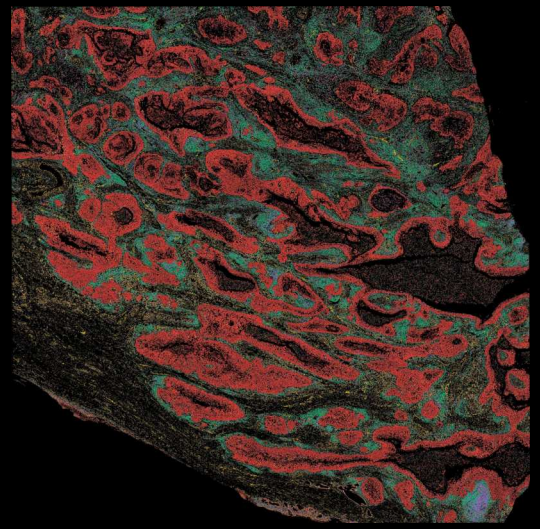


Datasets


144 datasets
62.3 million cells
580 samples
26 tissues
Genes per panel
Cells per dataset
Xenium (98)
CosMx (17)
MERFISH (13)
RAEFISH (3)
STARmap (3)
seqFISH (3)
osmFISH (2)
RIBOmap (1)
BARISTAseq (1)
EEL FISH (1)

Nicheverse can operate across imST platforms & biological settings

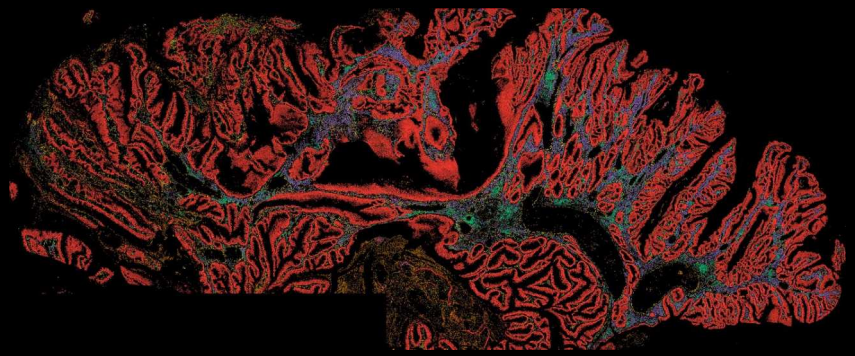
Human cervical cancer
Atera, 18,028 genes



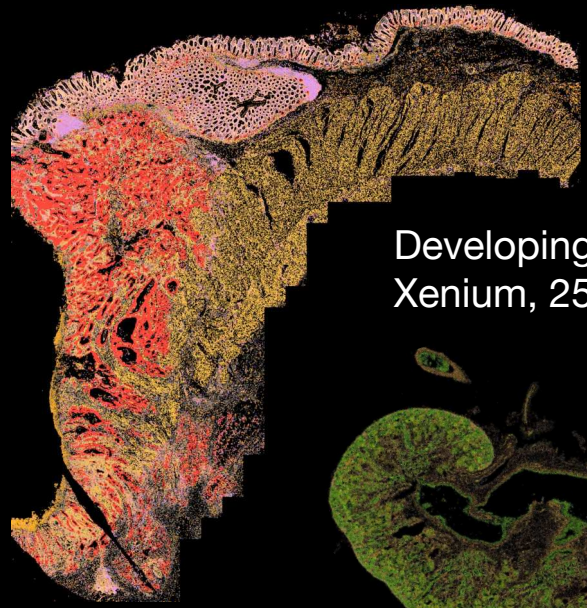
Mouse brain
Starmap+, 1,022 genes

Whole mouse pup
Xenium, 379 genes

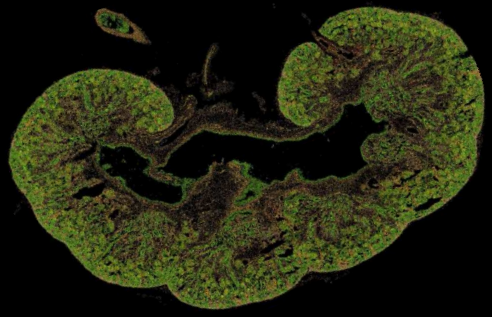
Human colorectal cancer, Xenium, 480 genes



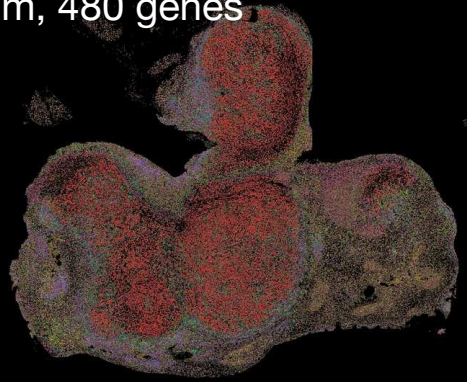
Human colon, CosMx, 18,935 genes



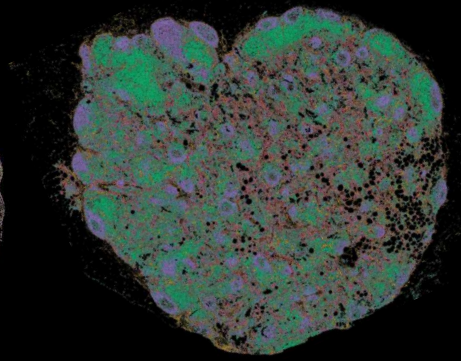
Developing mouse kidney
Xenium, 258 genes



Mouse PDAC (Perturb-map)
Xenium, 480 genes



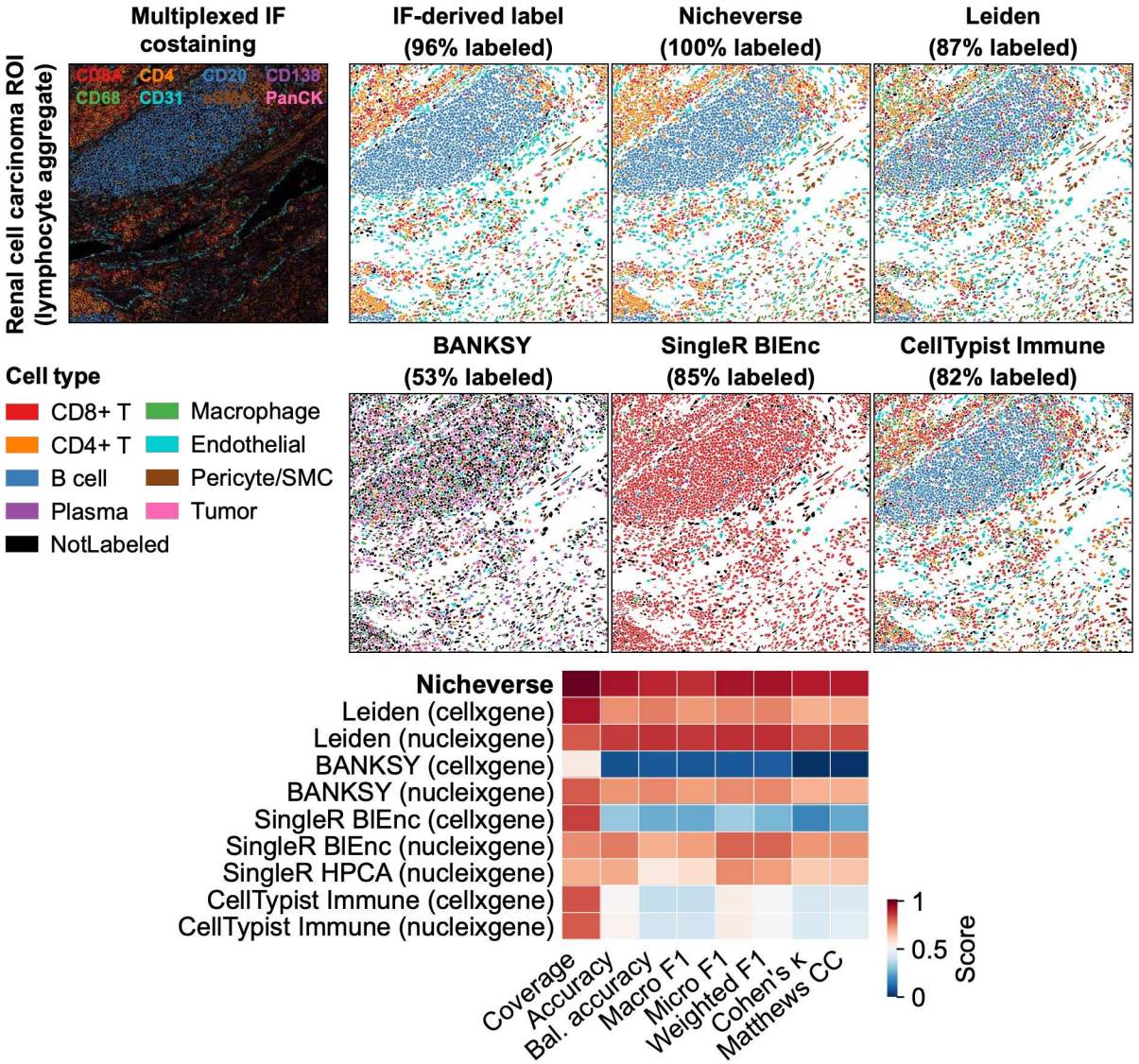
Human Lymph node
Xenium, 4,624 genes



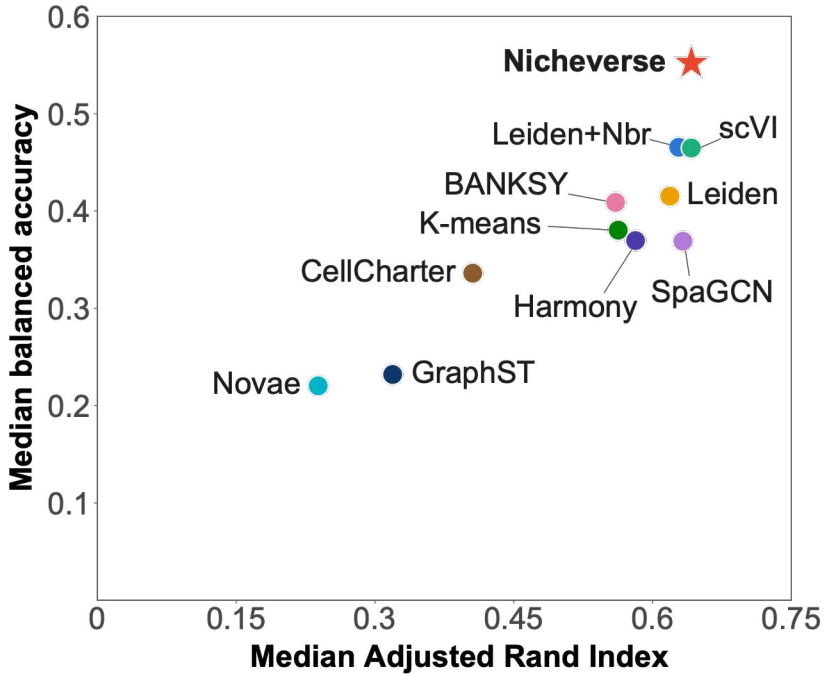
Benchmarks

How can we evaluate spatial representations when there is no single ground truth for tissue organization?

Concordance with protein costaining-derived annotations

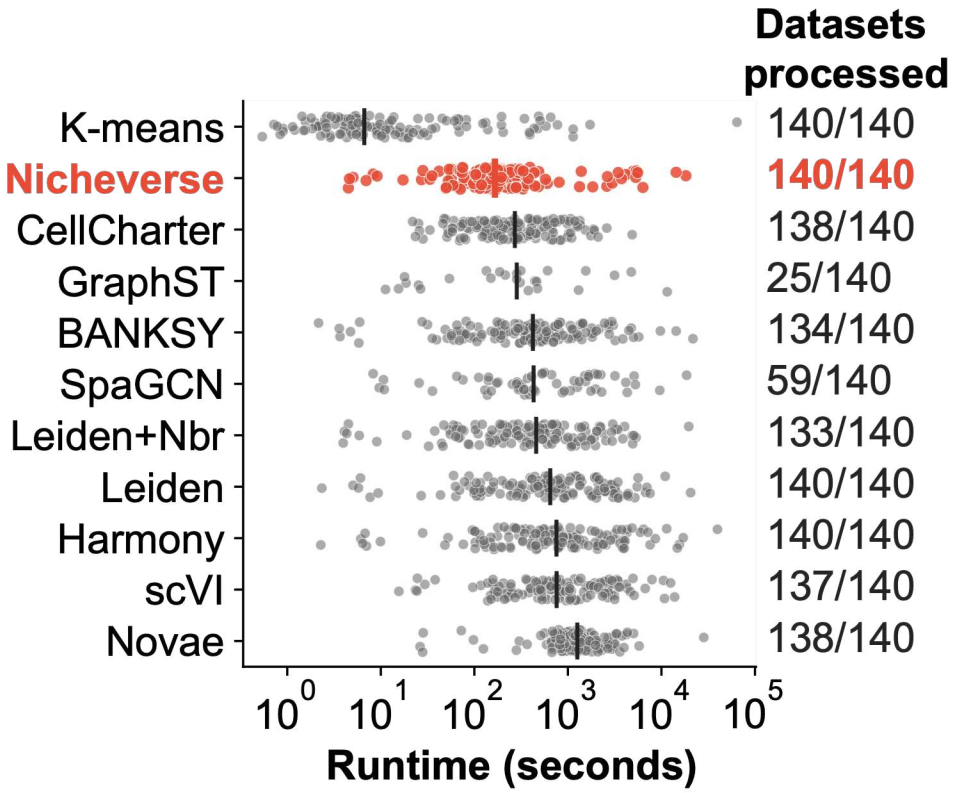


Concordance with reference annotations

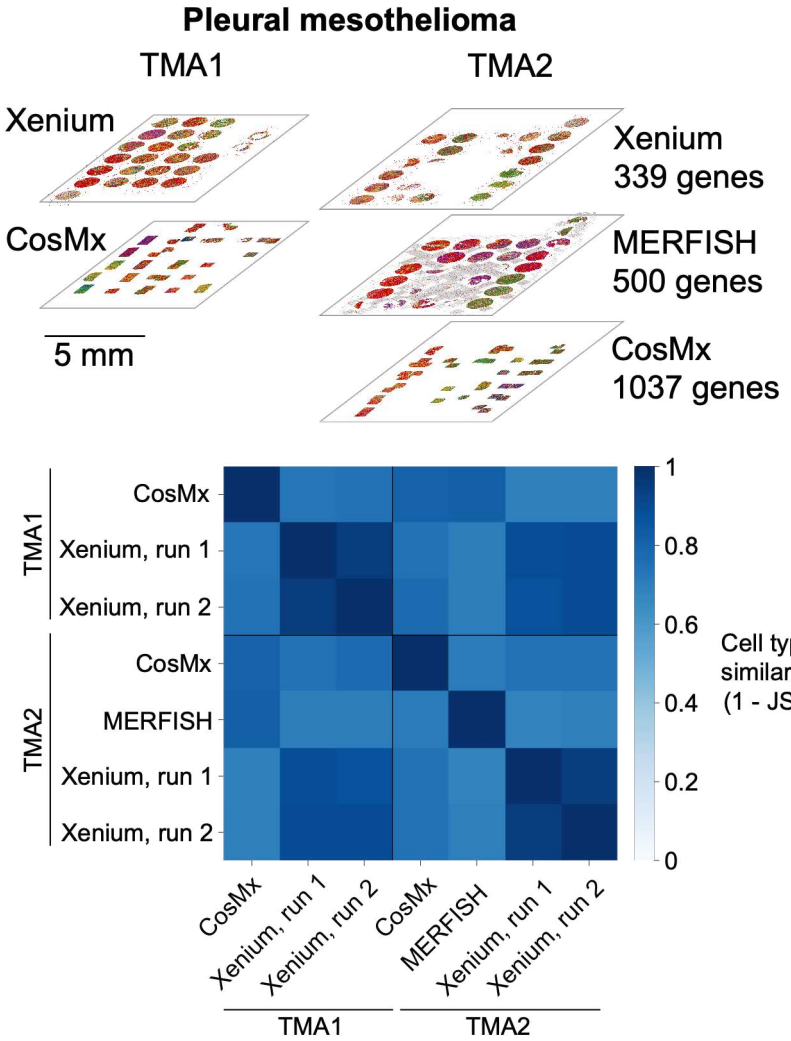


Benchmarks

Transcript-level modeling remains scalable with tractable compute



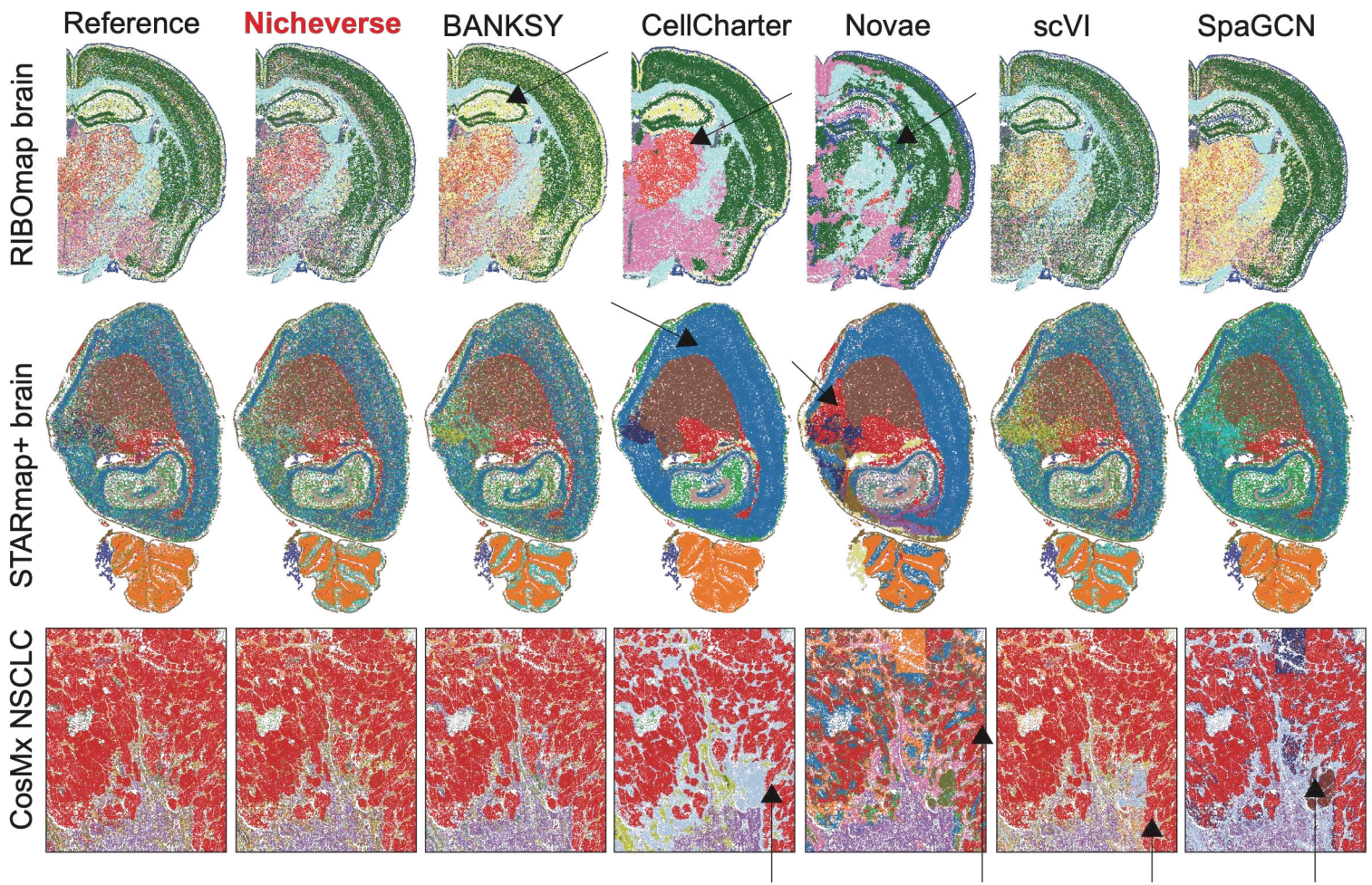
Reproducible across serial sections



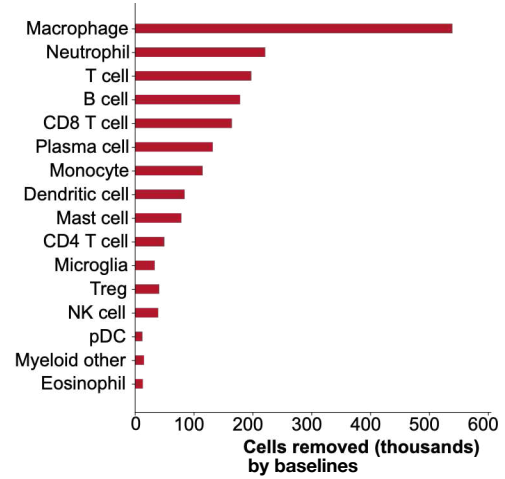
Dataset from Ozirmak et al., Nat. Comm. 2025

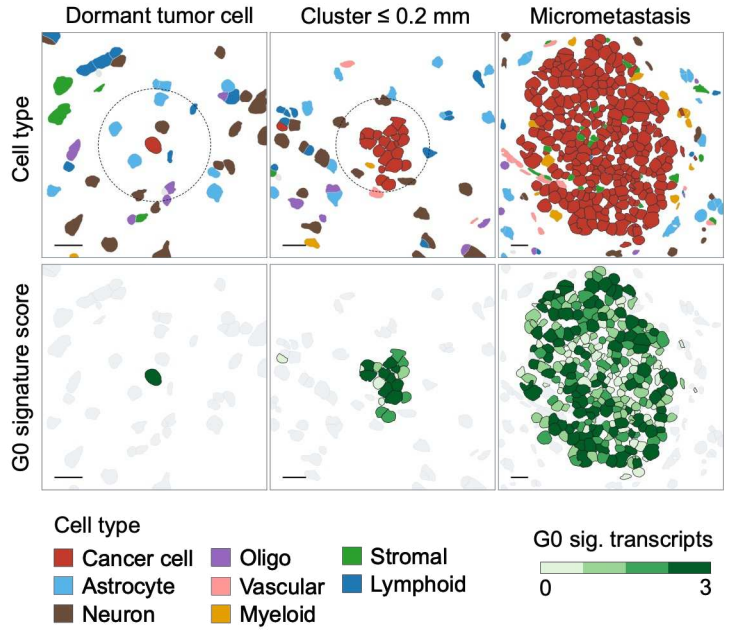
Benchmarks

Nicheverse avoids spatial oversmoothing



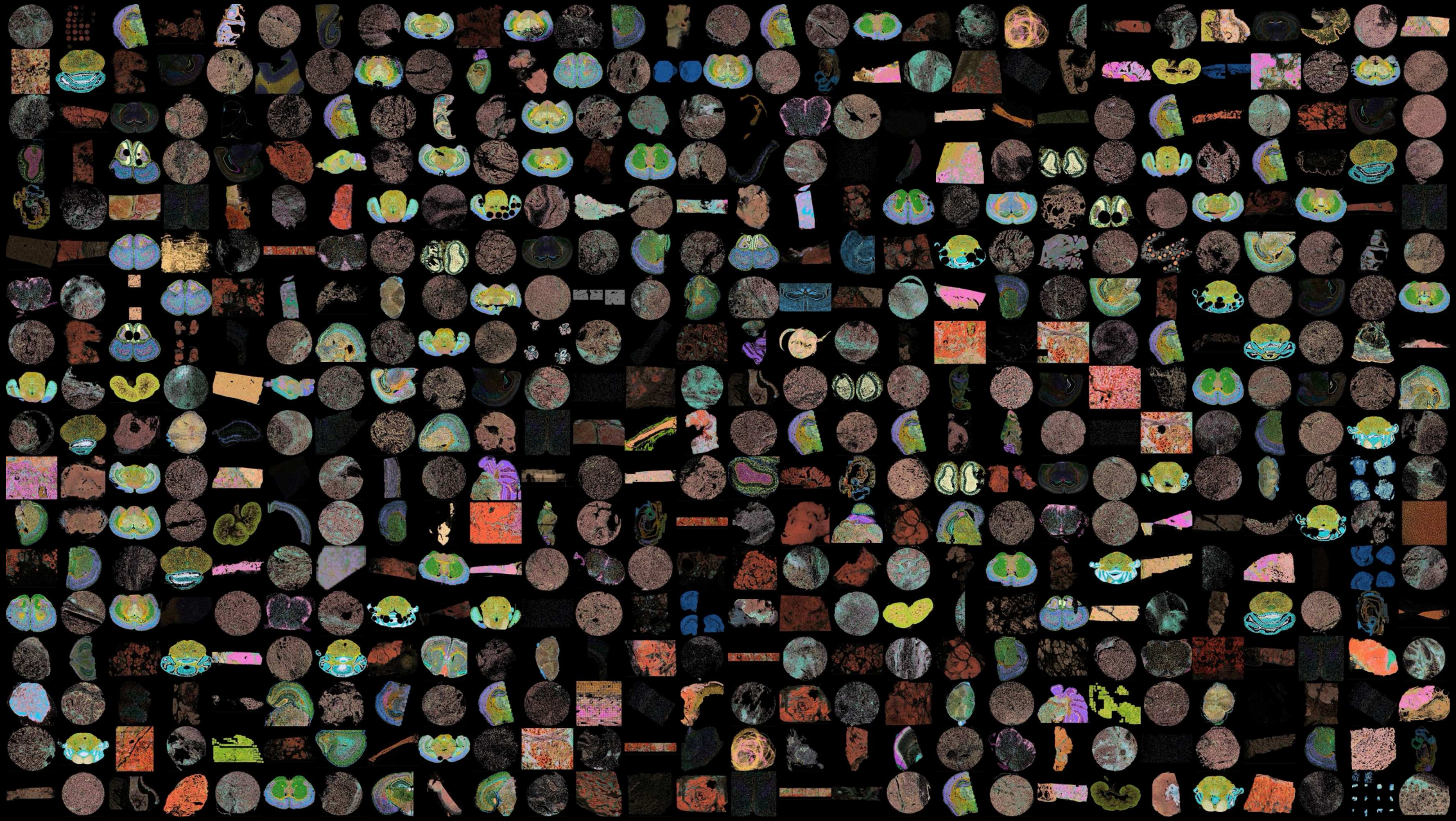
Nicheverse retains rare populations



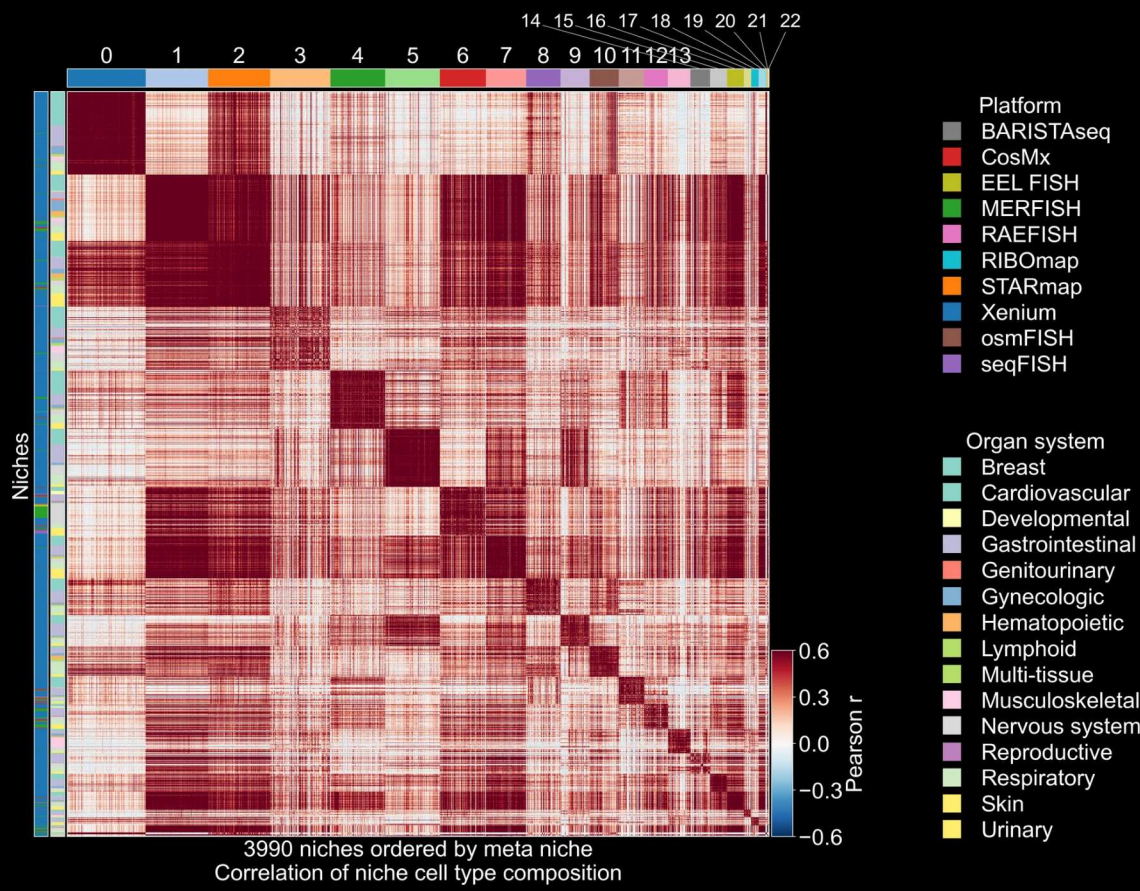


Breast-BrM Xenium dataset, Massagué & Gan labs

Are there multicellular niches that recur across biological contexts?



Nicheverse recovered recurring tissue environments across cancer, injury & normal tissue

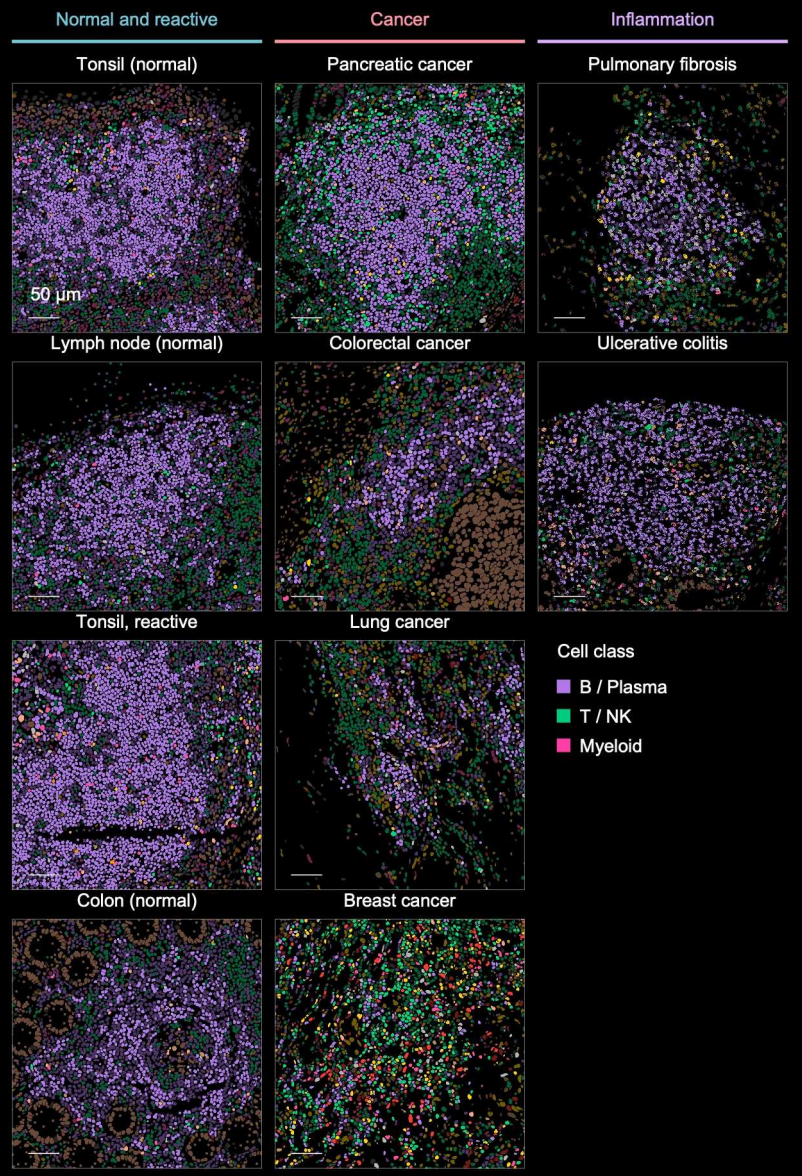


Platform
- BARISTAseq
- CosMx
- EEL FISH
- MERFISH
- RAEFISH
- RIBOmap
- STARmap
- Xenium
- osmFISH
- seqFISH

Organ system
- Breast
- Cardiovascular
- Developmental
- Gastrointestinal
- Genitourinary
- Gynecologic
- Hematopoietic
- Lymphoid
- Multi-tissue
- Musculoskeletal
- Nervous system
- Reproductive
- Respiratory
- Skin
- Urinary

Meta niche
- 0 Endothelial and plasma cell
- 1 Mixed epithelial and immune
- 2 Vascular immune
- 3 Neuronal and epithelial
- 4 Smooth muscle
- 5 Macrophage rich
- 6 Neuronal parenchyma
- 7 Macrophage and fibroblast stroma
- 8 T and B lymphocyte
- 9 Dendritic cell and macrophage
- 10 B cell aggregate
- 11 Plasma cell
- 12 Astrocyte like
- 13 Erythroid
- 14 Hepatocyte and proximal tubule
- 15 Pericyte perivascular
- 16 Smooth muscle and epithelial
- 17 Neutrophil and dendritic cell
- 18 Mast cell
- 19 Sparse tumor microenvironment
- 20 Capillary endothelium
- 21 Stromal
- 22 Brain macrophage

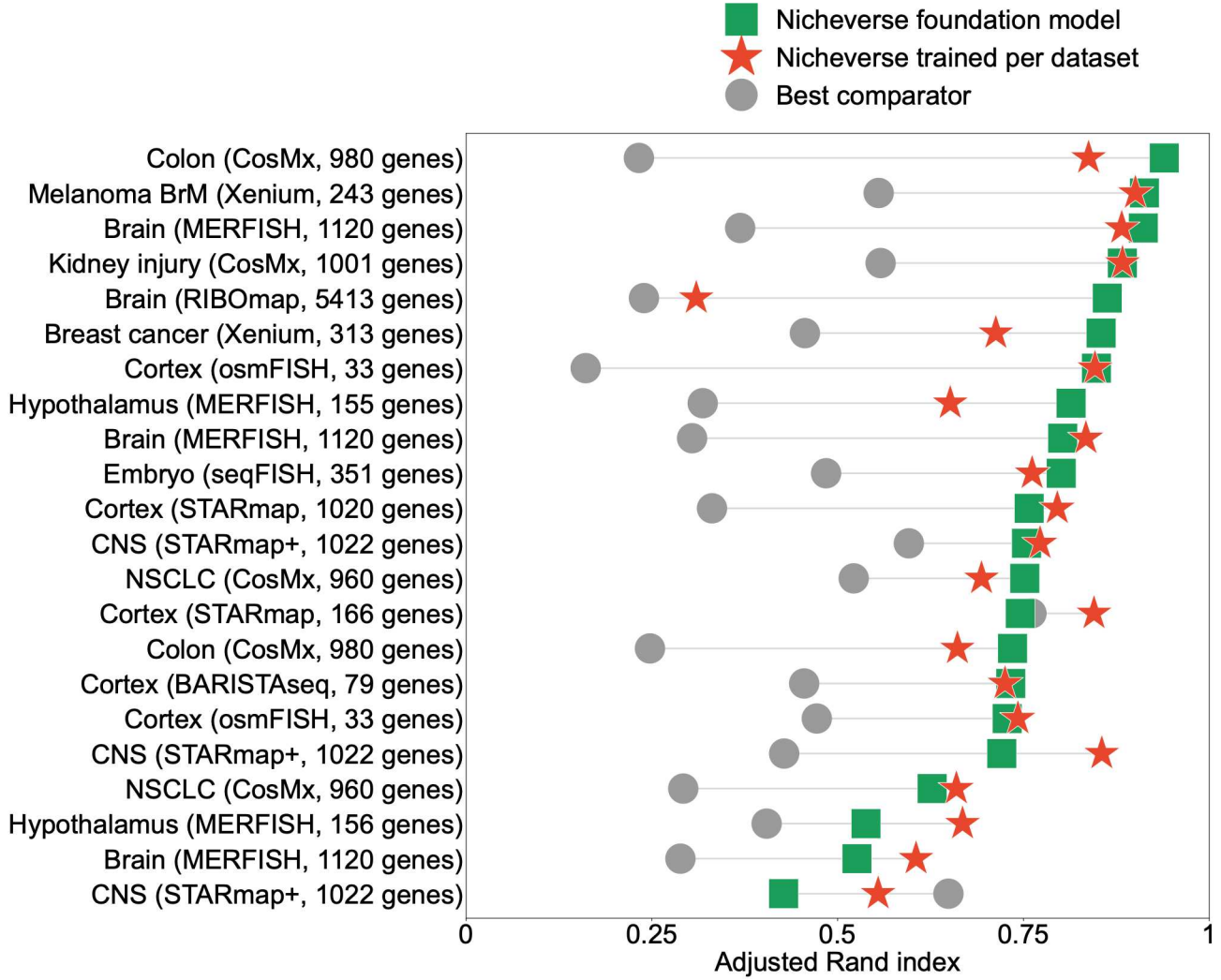
B & T cell aggregates



Shared pretraining enables transferable representations across heterogeneous spatial datasets

Can we pretrain a single, large model to learn shared representations between datasets?

Need more diverse spatial datasets!



Summary: modeling of imST data with Nicheverse

- Nicheverse enables **scalable, interpretable modeling of imST data** and can be extended as **general-purpose framework** to model transcriptomics, proteomics and histopathology data.

- Limitations
  - Need raw data (transcript coordinates). Can operate on cell by gene matrices but cannot denoise segmentation errors pos-hoc.
  - Works with seqST data but cannot handle challenges specific to seqST platforms.

- Future directions
  - Improve accuracy & reliability of agentic-annotation.
  - Incorporate probe-level sensitivity & specificity into the observation model.

- https://nicheverse.org/
- http://github.com/digvijaky/nicheverse

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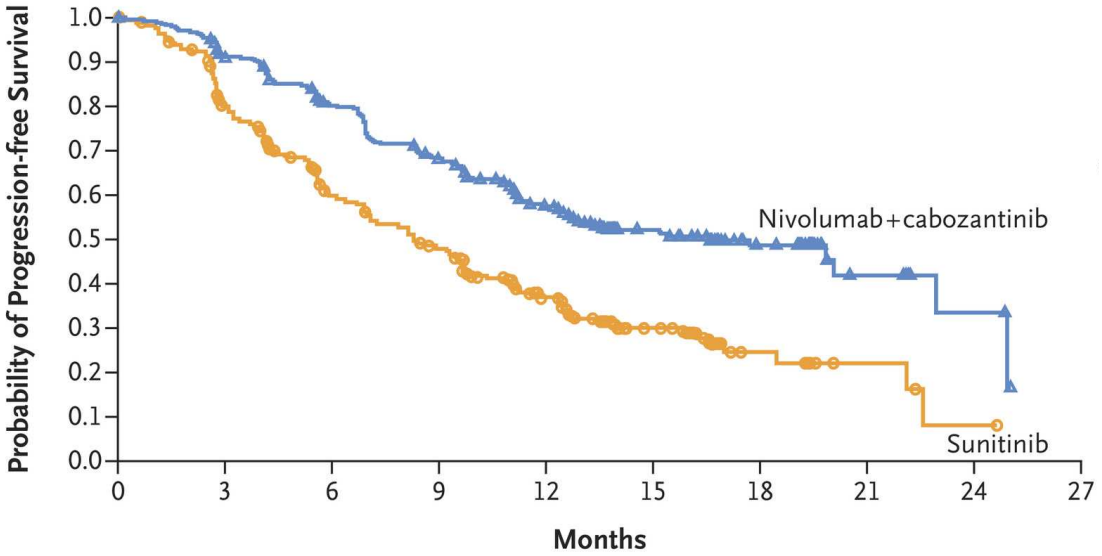
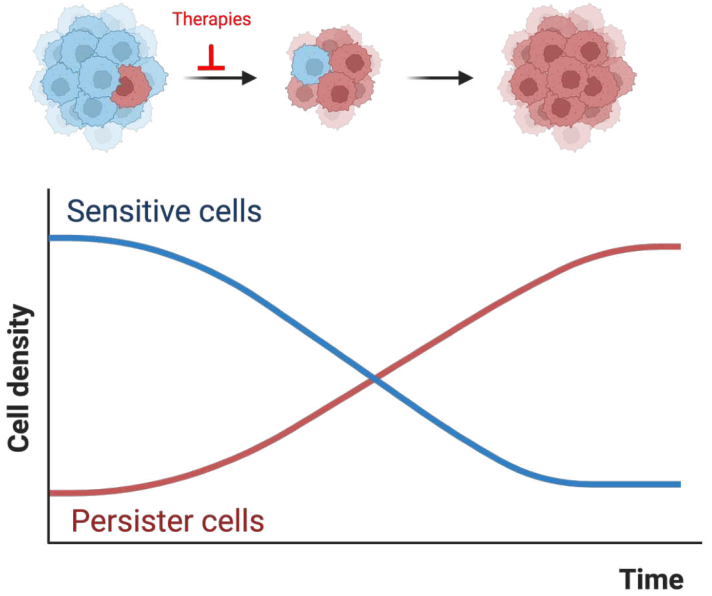
Resistance to immunotherapy is a major clinical challenge in kidney cancer

Clear cell renal cell carcinoma (ccRCC) tumors initially respond well to frontline immune checkpoint inhibitor therapies (ICIs), but many patients experience relapse over time.



Choueiri et al. NEJM 2021

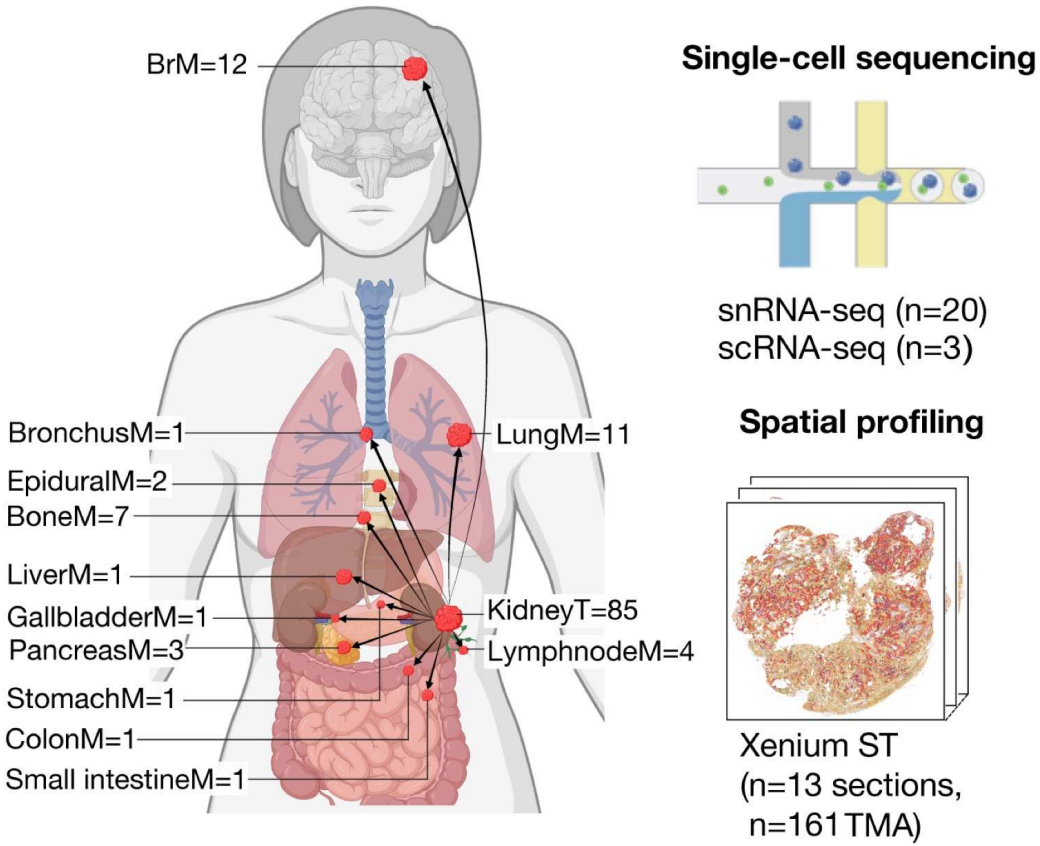


- Prior studies associated ICI response with various cell types in the TME. (Krishna et al. *Cancer Cell* 2021).
- Cancer-cell phenotypes driving resistance remain less well-defined.

What is the phenotype of cancer cells that reinitiate tumor growth after checkpoint blockade in ccRCC?

Tumor heterogeneity in LUAD ~ continuum of developmental states (Laughney et al., Nat. Med. 2020)

A multi-site single-cell & spatial atlas of ccRCC tumors



Single-cell sequencing

snRNA-seq (n=20)
scRNA-seq (n=3)

Spatial profiling

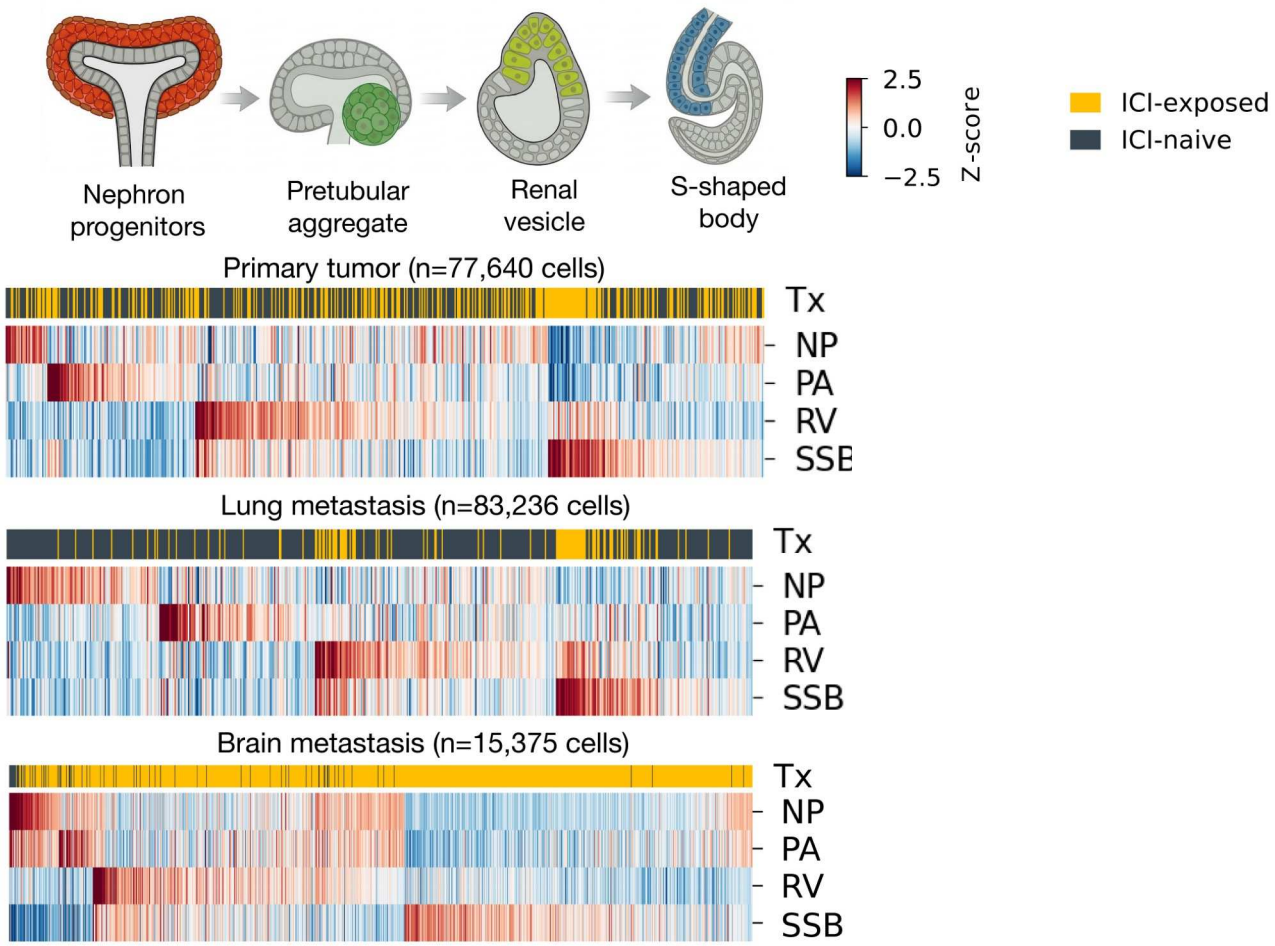
Xenium ST
(n=13 sections,
n=161 TMA)

- Multi-modal profiling of primary kidney tumors and metastases from 11 anatomic sites across 99 patients.
- Largest single-cell-profiled cohort of ccRCC metastases to date

Do ccRCC cells engage programs along the nephrogenic developmental axis?

Cohort assembled with Ari Hakimi, Yingbei Chen, Ritesh Kotecha, Andrea Lopez, Hui Jiang, Lynda Vuong

ccRCC cells display a continuum of nephrogenic developmental states



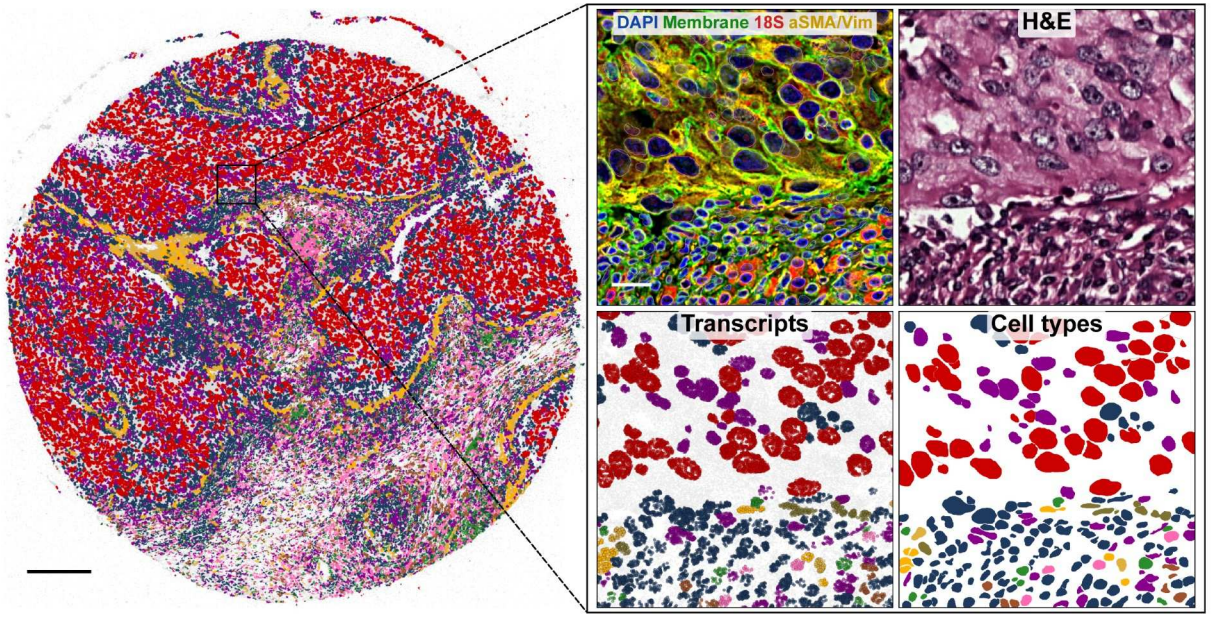
Committed developmental states are enriched in metastatic ccRCC.

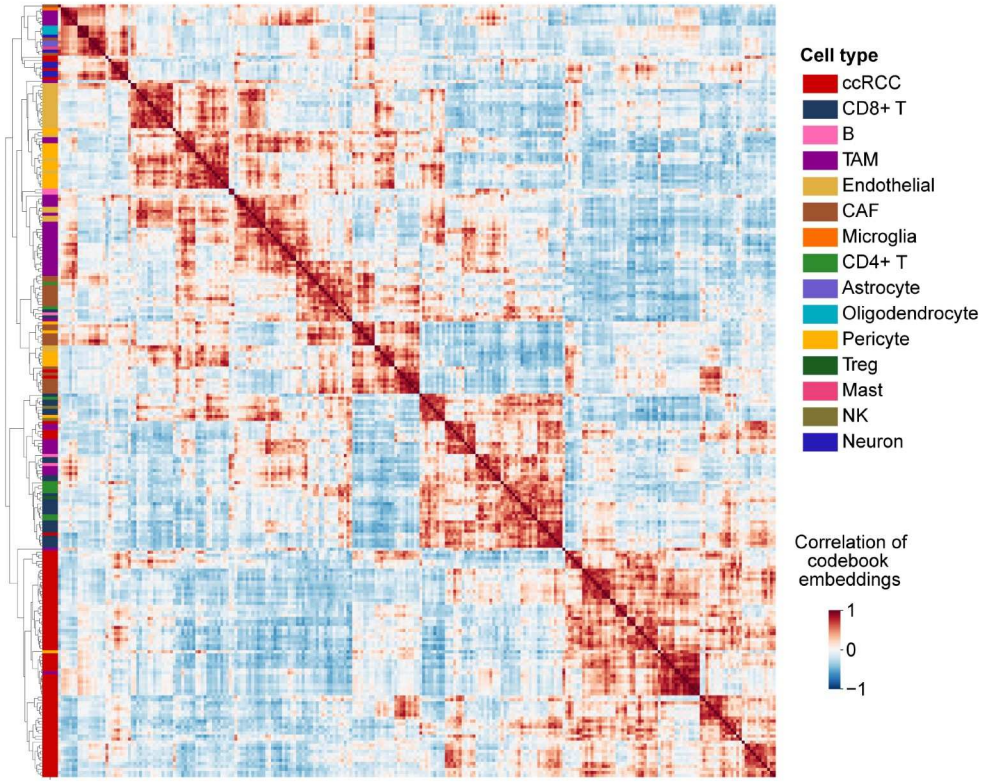
RCC cancer cells shift toward commitment to a developmental trajectory after ICI exposure.

How do these cell states interact with the TME?

Nicheverse maps cell states & spatial niches across the ccRCC atlas

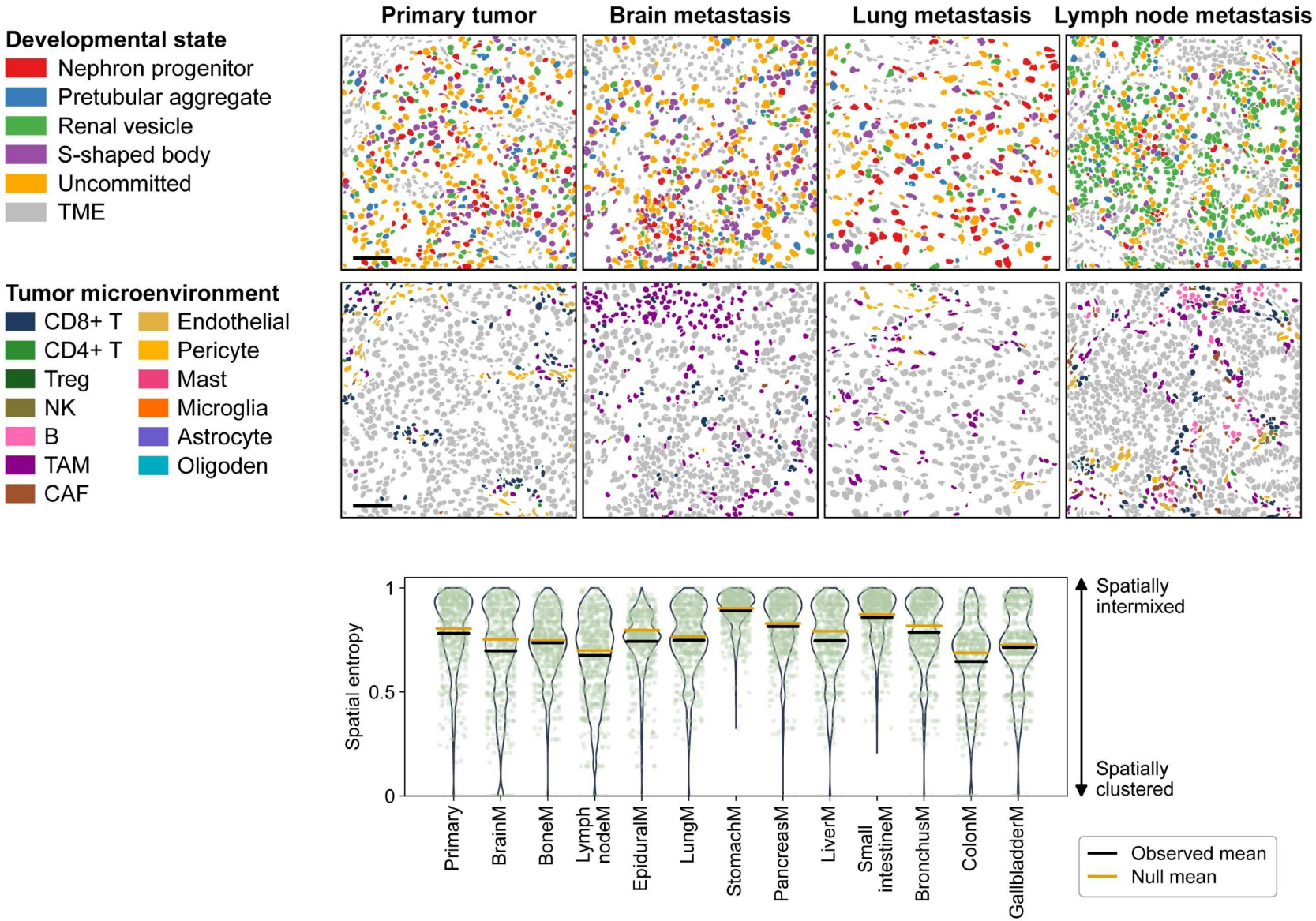
110 patients
Primary tumors & 11 metastatic sites
Xenium ST (366-gene panel)



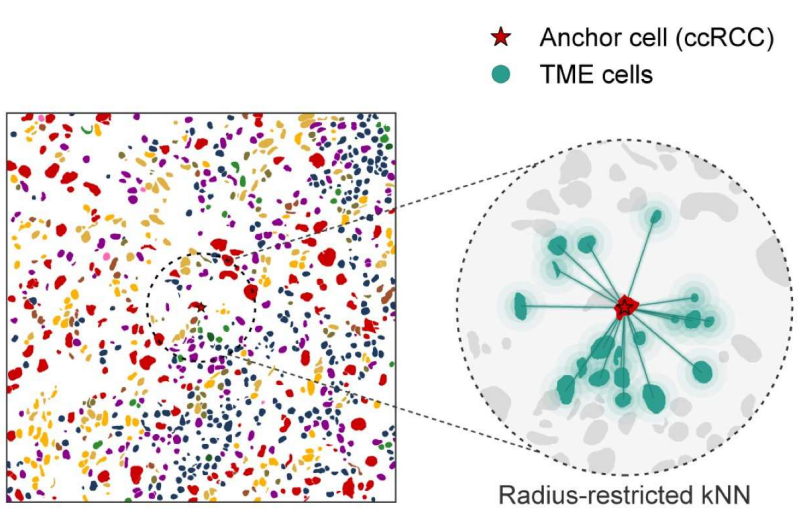


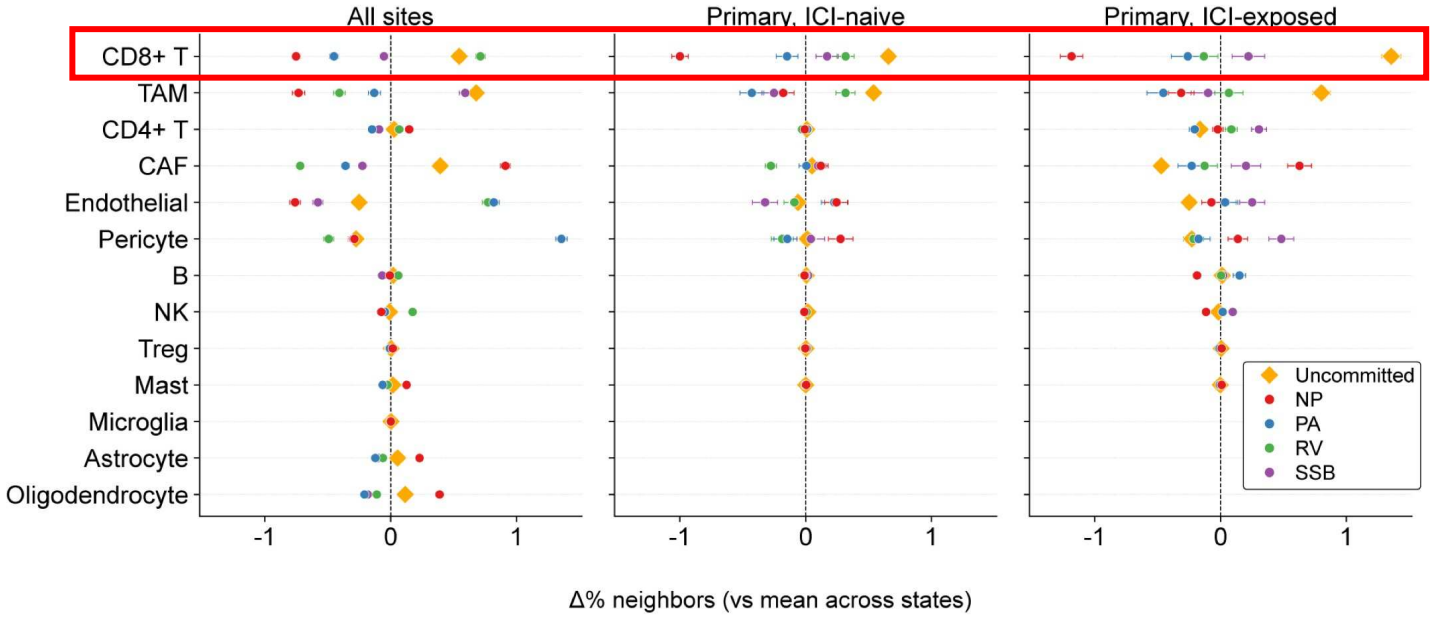
Hierarchical clustering of cell type codebook embeddings

Cancer cells in various developmental states spatially coexist within tumors

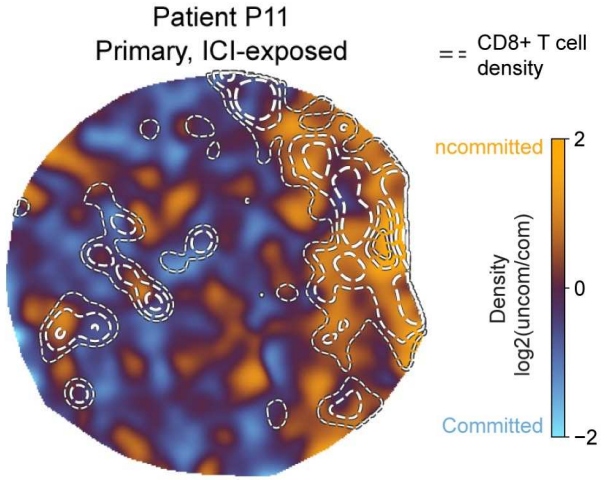


Commitment to developmental trajectory may confer immune evasive advantage





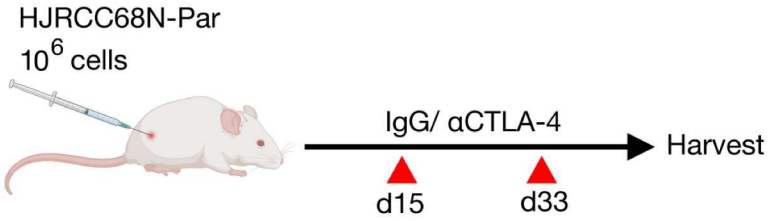
Committed cancer cells are spatially segregated from immune infiltrates.

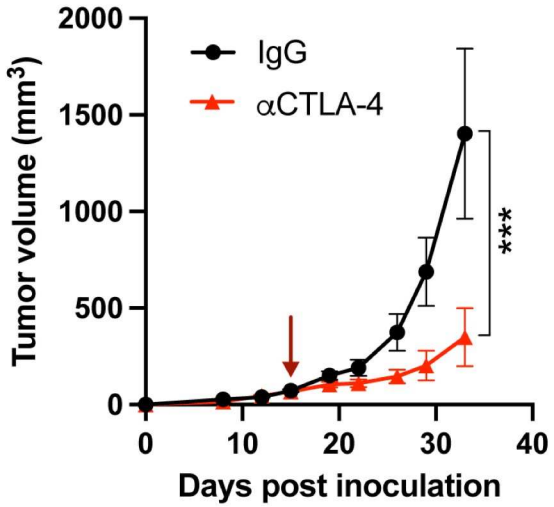


Do immunotherapy actively select for or induce this developmental commitment of cancer cells?

For functional validation, we need immunocompetent preclinical mouse models, which can generate metastasis & also respond to immune checkpoint blockade therapies.

HJRCC68N provides an immunocompetent model of aCtla-4-responsive ccRCC tumors

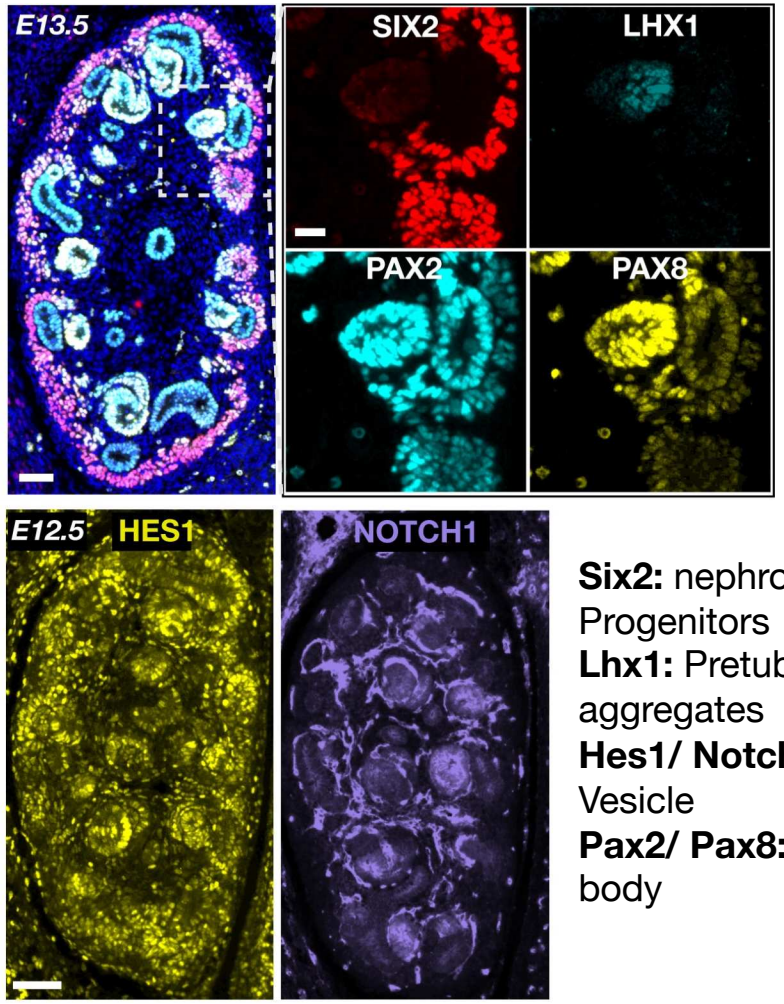




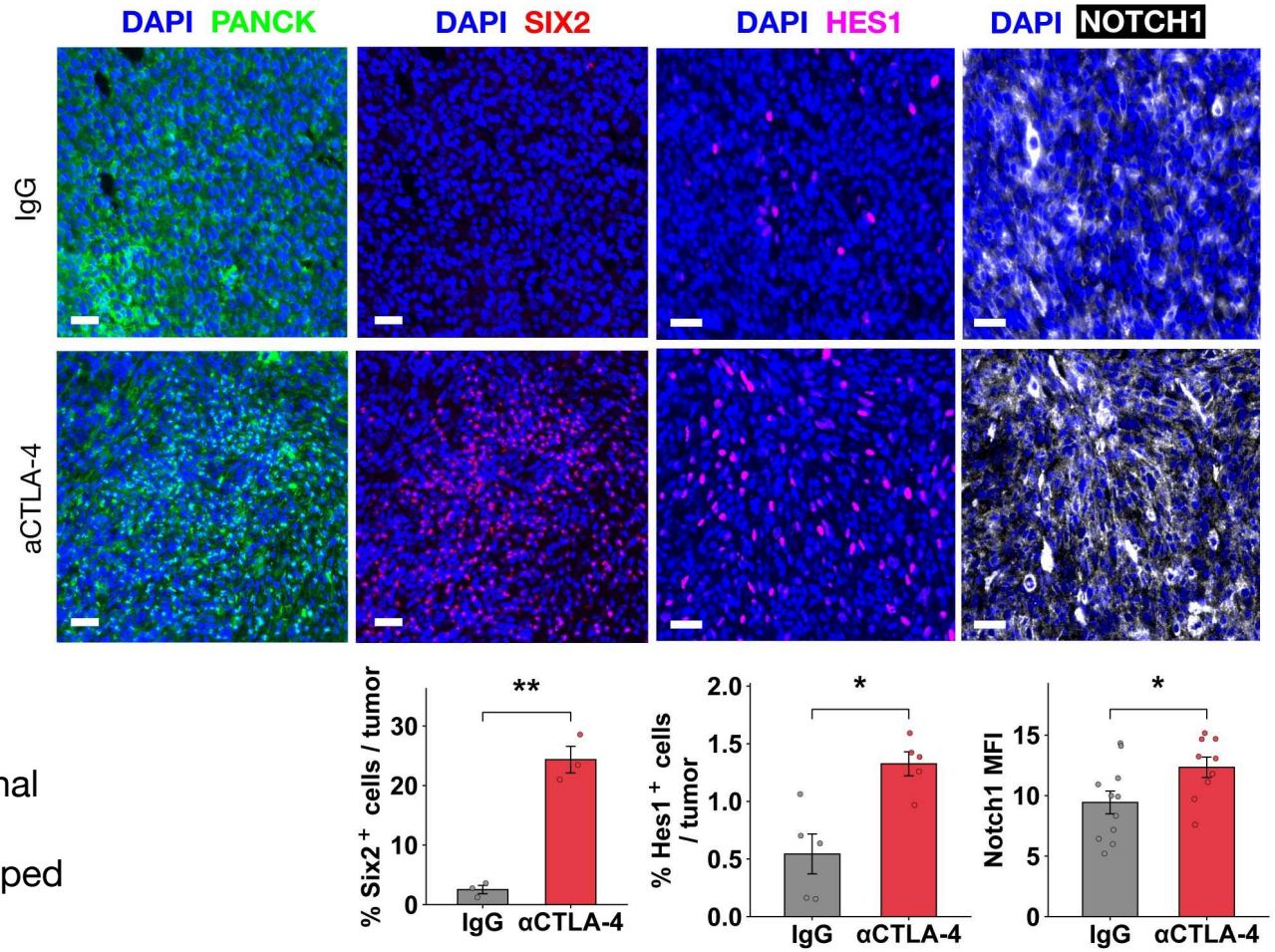
HJRCC68N cell line developed
by Hui Jiang, Hakimi lab

Does αCTLA-4 selectively enrich for nephrogenic developmental states?

Nephron progenitor-like cancer cells are enriched after αCTLA-4 exposure

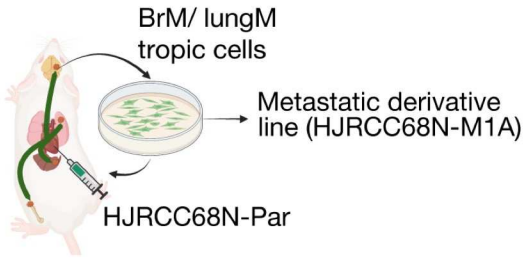


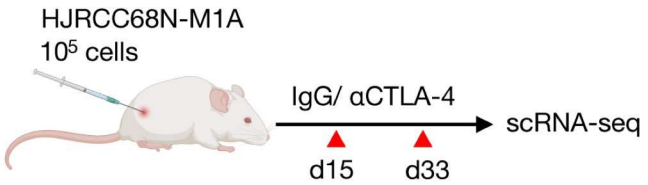
Six2: nephron Progenitors
Lhx1: Pretubular aggregates
Hes1/ Notch1: Renal Vesicle
Pax2/ Pax8: S-shaped body

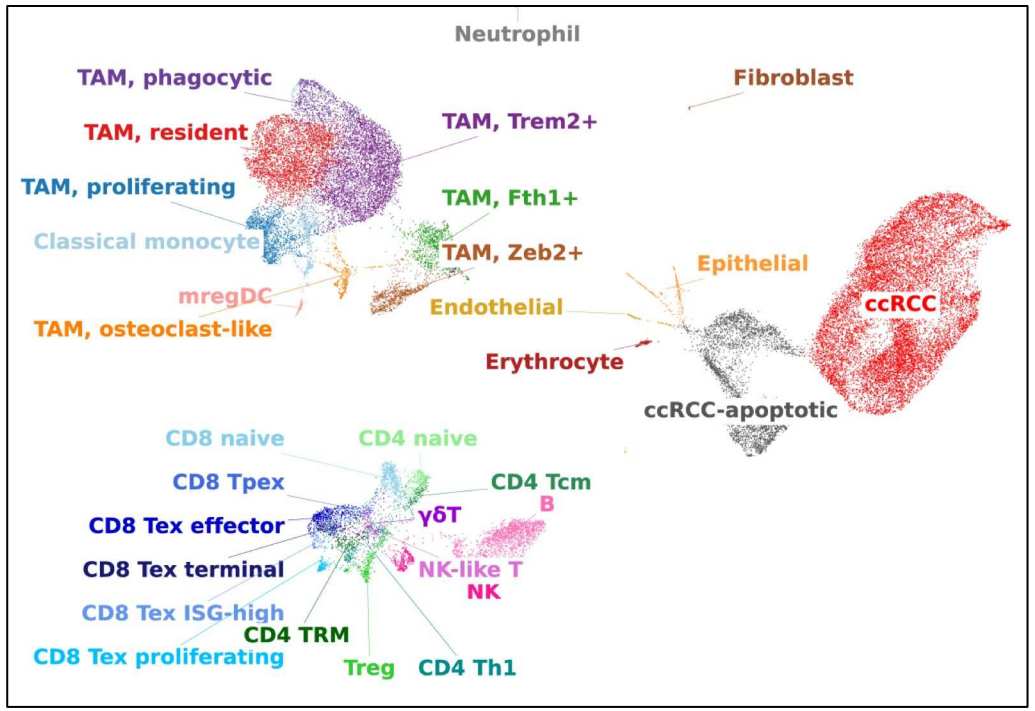


Is this phenotype conserved in metastatic setting?

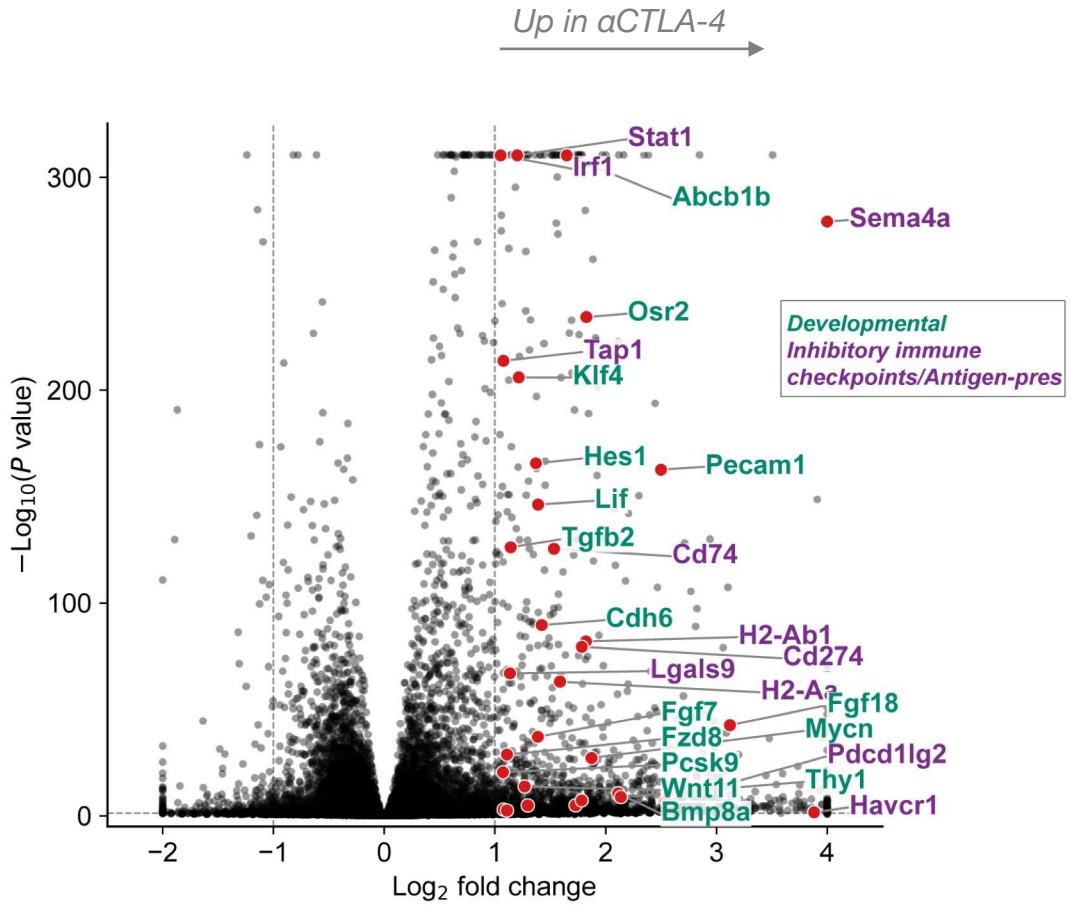
Metastatic derivatives of HJRCC68N respond to aCTLA-4 in orthotopic setting

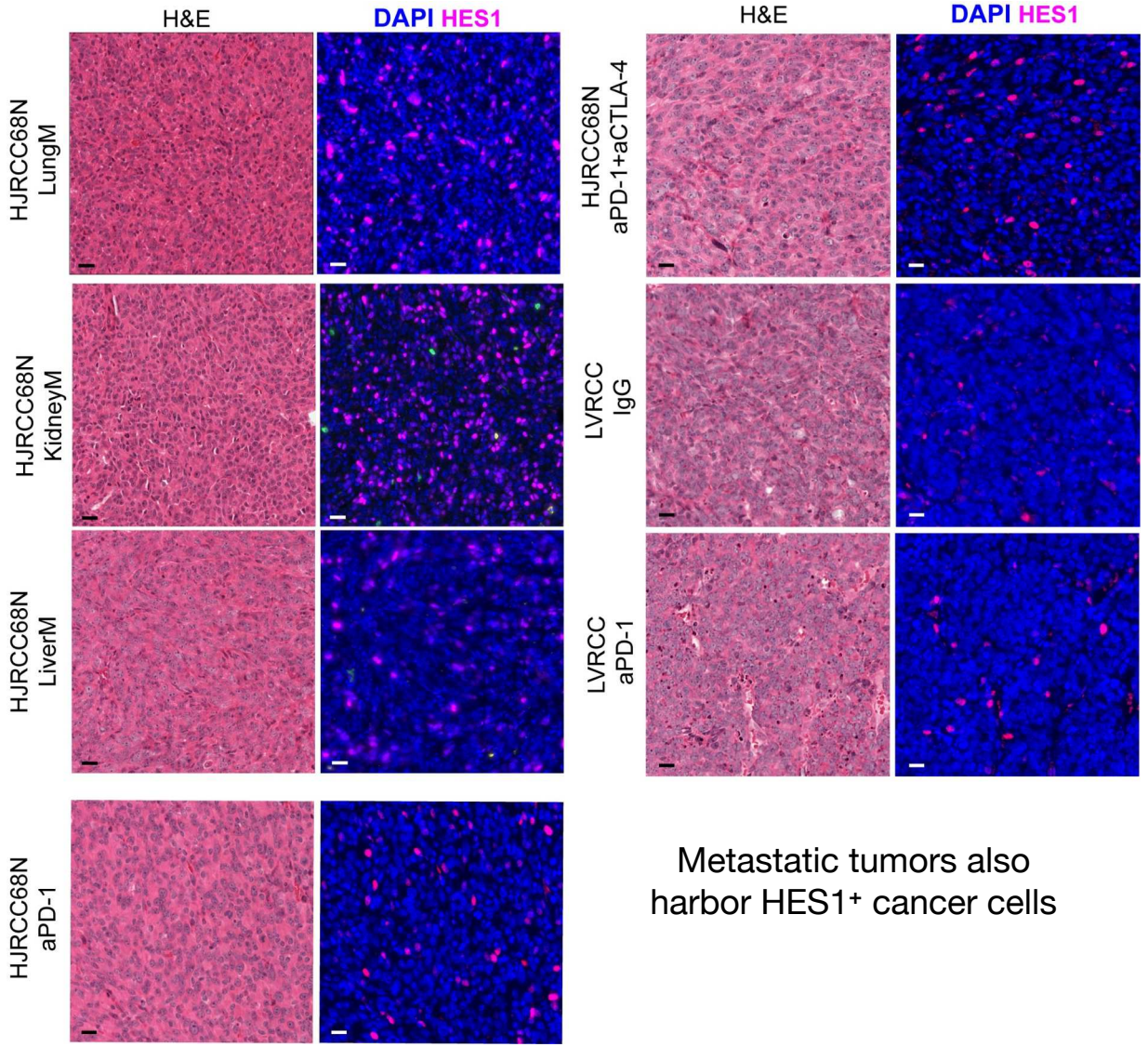






Persister cells overexpress nephrogenic programs & inhibitory immune checkpoints

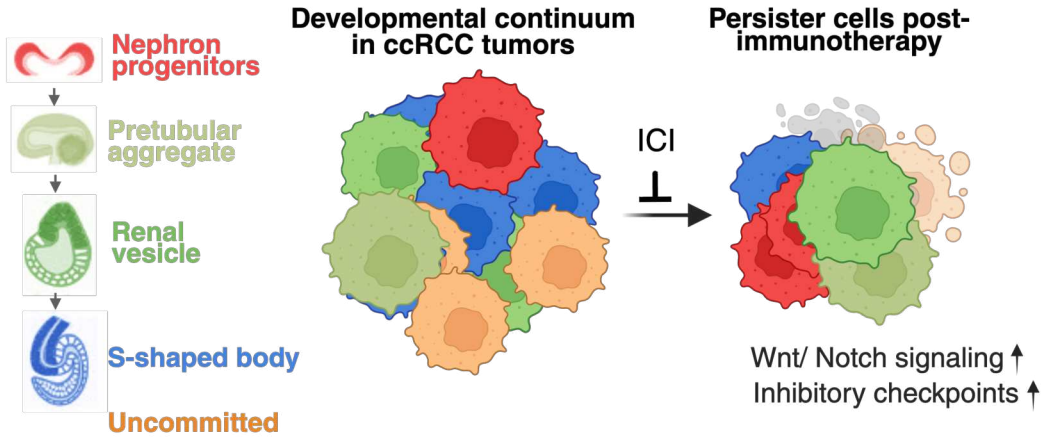




Metastatic tumors also harbor HES1⁺ cancer cells

Summary 2: nephrogenic developmental states are associated with ICI persistence in ccRCC

- ccRCC cancer cells adopt nephrogenic developmental programs and developmentally committed states are enriched following immune checkpoint blockade.
- Future combinational therapies may need to re-sensitize these cancer cells through alternative inhibitory immune checkpoints.
- Limitations
  - A direct mechanistic link between HES1 activity, inhibitory checkpoint expression & ICI persistence is not established.
  - Currently available preclinical models of ccRCC limit our ability to evaluate the same treatment combinations used in clinical practice.



Outline

Chapter 1. Nicheverse: Modeling Spatial Architecture Using Discrete Representation Learning

Yarlagadda D, Massagué J, Leslie C. *Discrete representation learning for modeling imaging based spatial transcriptomics data. Proceedings of the ICCV Workshop on Bioimage Computing*. 2023;3848–3857.

Yarlagadda D, Leslie C. *CellSplat: End to End 3D Gaussian Splatting for Cellular Decomposition of Spatial Transcriptomics. CVPR Workshop on End-to-End 3D Learning*. 2026.

Yarlagadda D, Massagué J, Leslie C. *Nicheverse: discrete representation learning framework for modeling spatial transcriptomics data*. Manuscript in preparation. 2026.

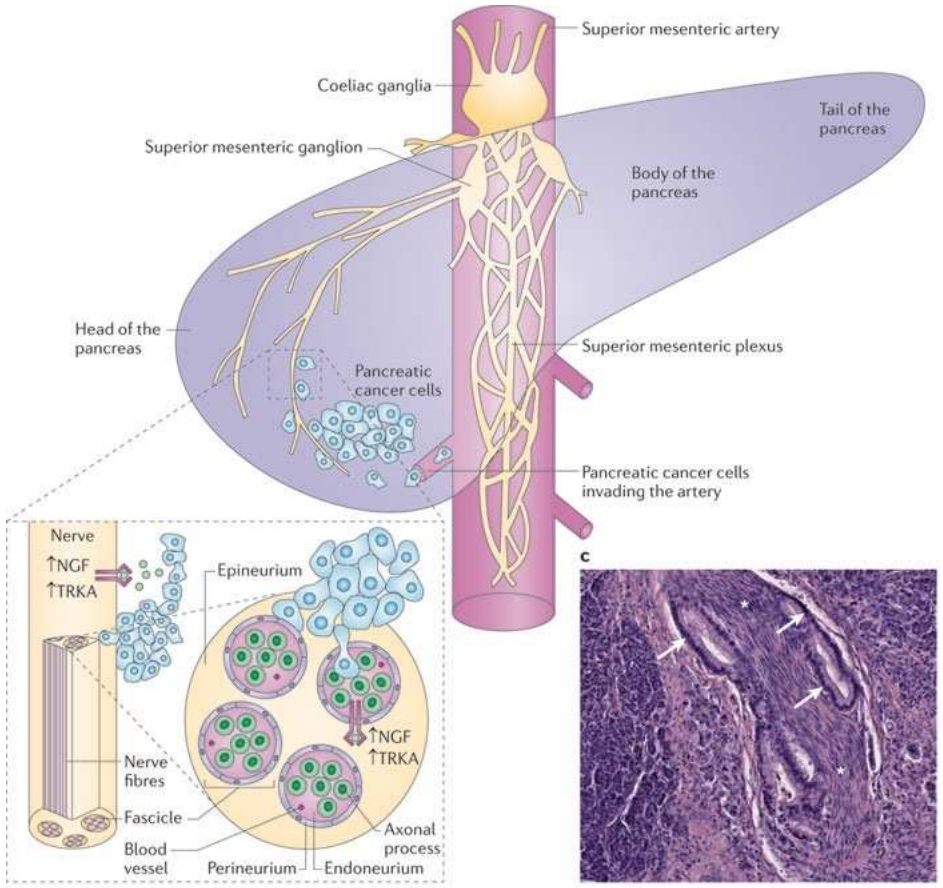
Chapter 2. Developmental Reversion Underlies Resistance to Immune Checkpoint Blockade in Kidney Cancer

Yarlagadda D, Wang Z, Jiang H, Vuong L, López Sanmiguel A, Yang C, Kotecha R, Chen Y, Hakimi A, Leslie C, Massagué J. *Developmental reversion underlies resistance to immune checkpoint blockade in kidney cancer. bioRxiv*. 2026.

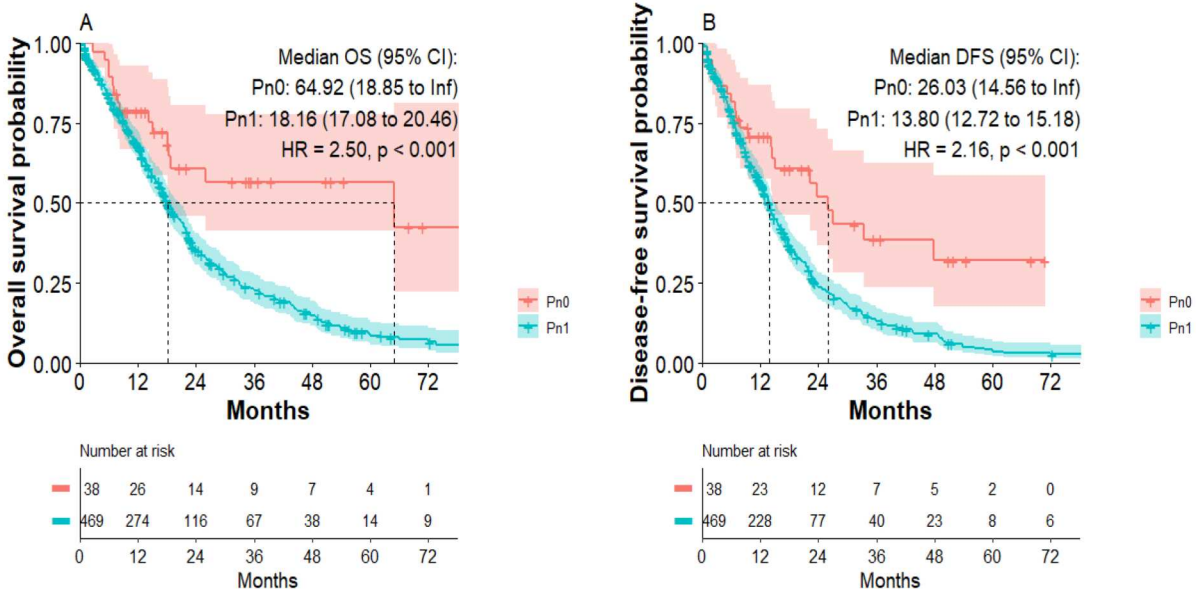
Chapter 3. Invasion Status Stratifies the Composition of the Human Pancreatic Cancer Perineural Niche

Hussain Z*, **Yarlagadda D***, Li Q, Tezcan N, Stupakov P, Sadatrezaei G, Wong R, Massagué J, Tarcan Z, Basturk O, Deborde S, Leslie C, Sherman M. *Invasion status stratifies the composition of the human pancreatic cancer perineural niche*. *bioRxiv*. 2026.

Perineural invasion in pancreatic ductal adenocarcinoma



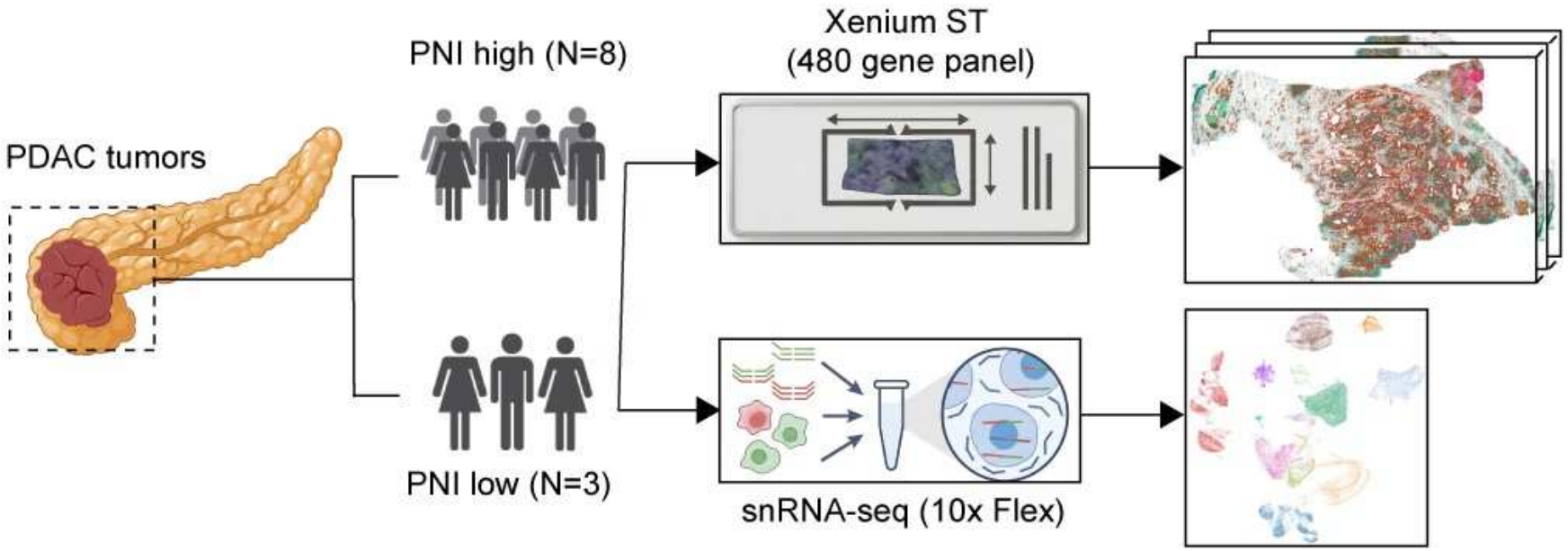
Bapat et al. *Nat. Rev. Cancer* 2011



Felsenstein *et al*. 2022

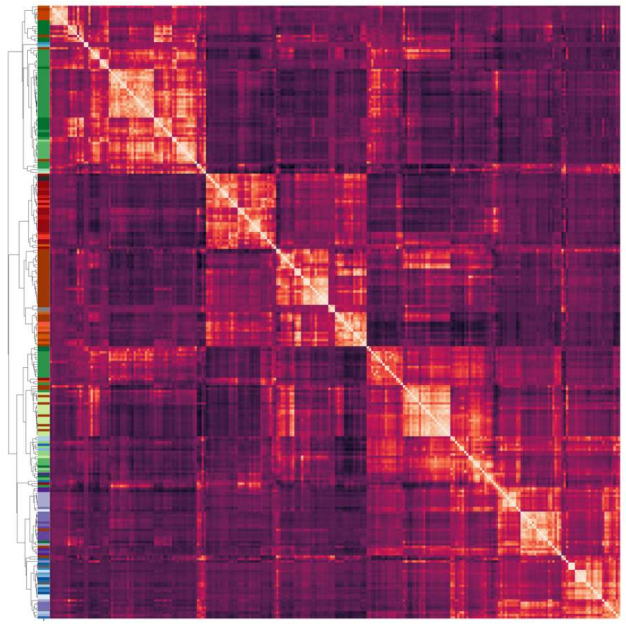
How does perineural invasion reshape the local tumor microenvironment in PDAC?

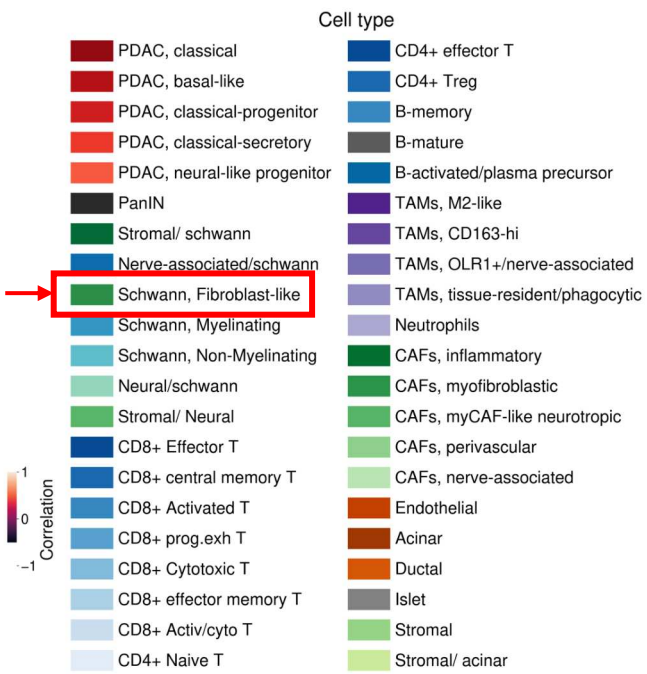
Spatial profiling of PNI^{+} vs PNI^{-} PDAC tumors



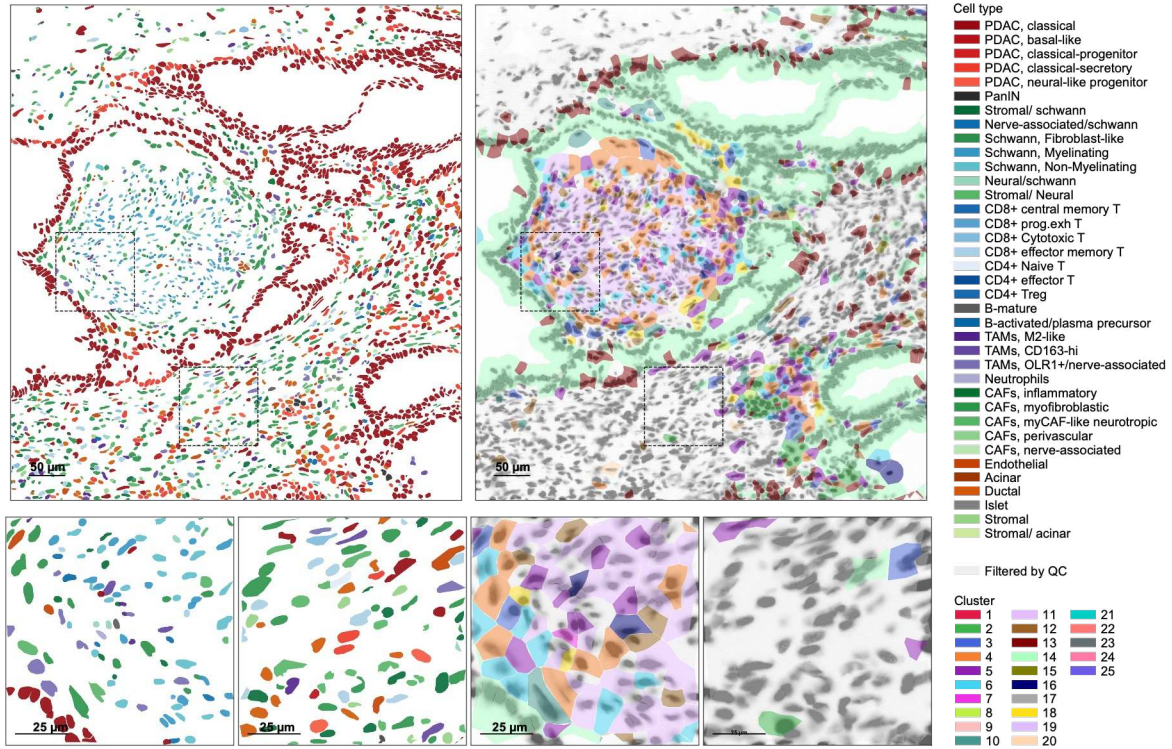
Collaboration between Leslie lab (Qianzi), Sherman lab (Zainab) & Deborde lab

Nicheverse recovered fine-grained cell states in PDAC tumors

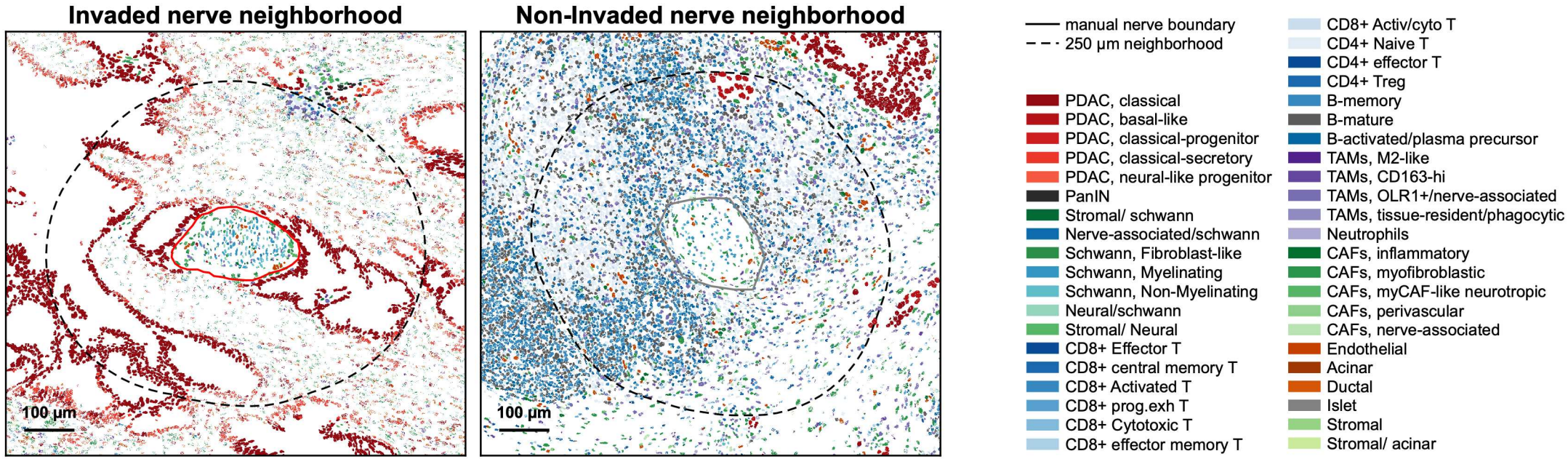




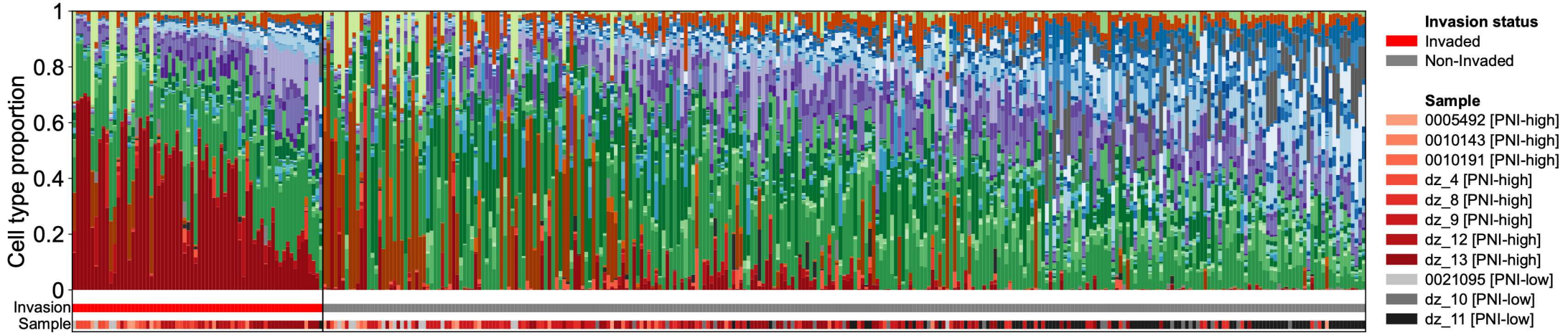
Nicheverse annotation *10x segmentation + filtering*



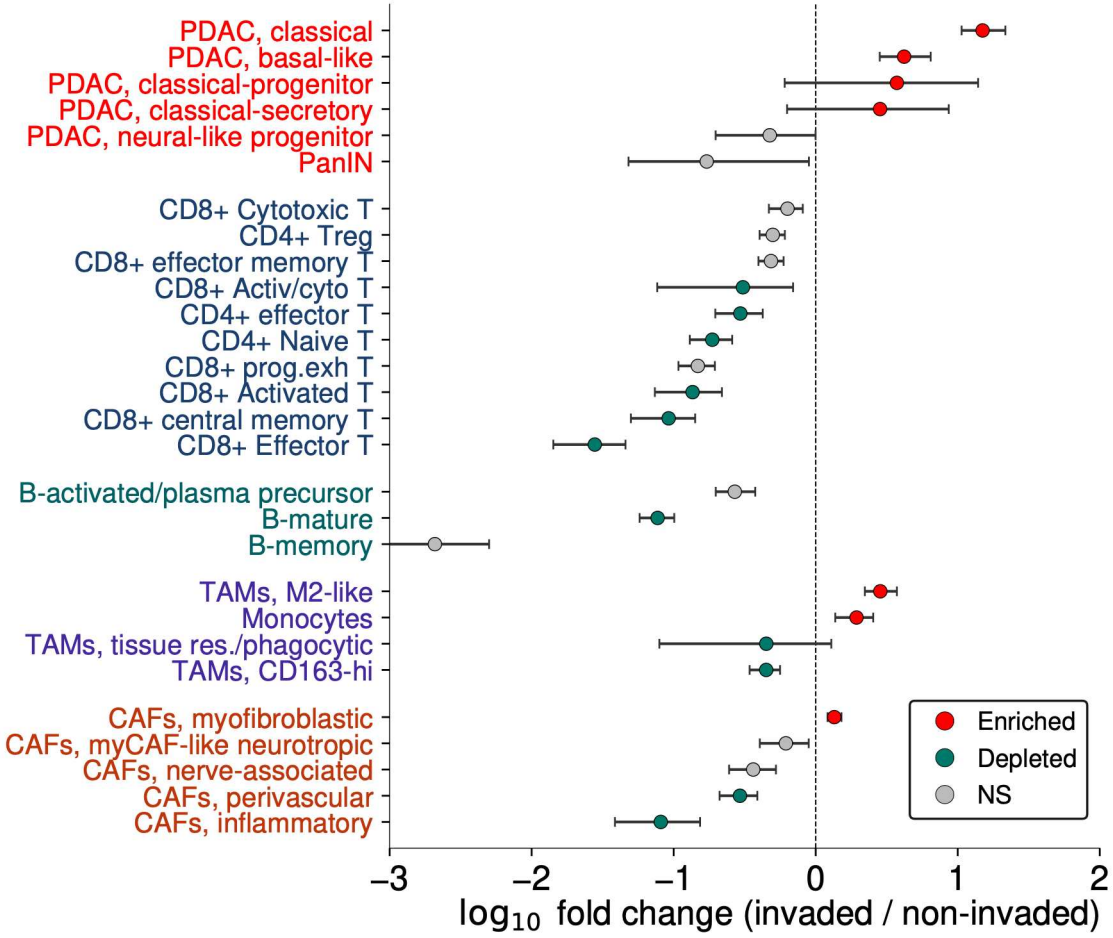
Invasion status stratifies the composition of the human pancreatic cancer perineural niche

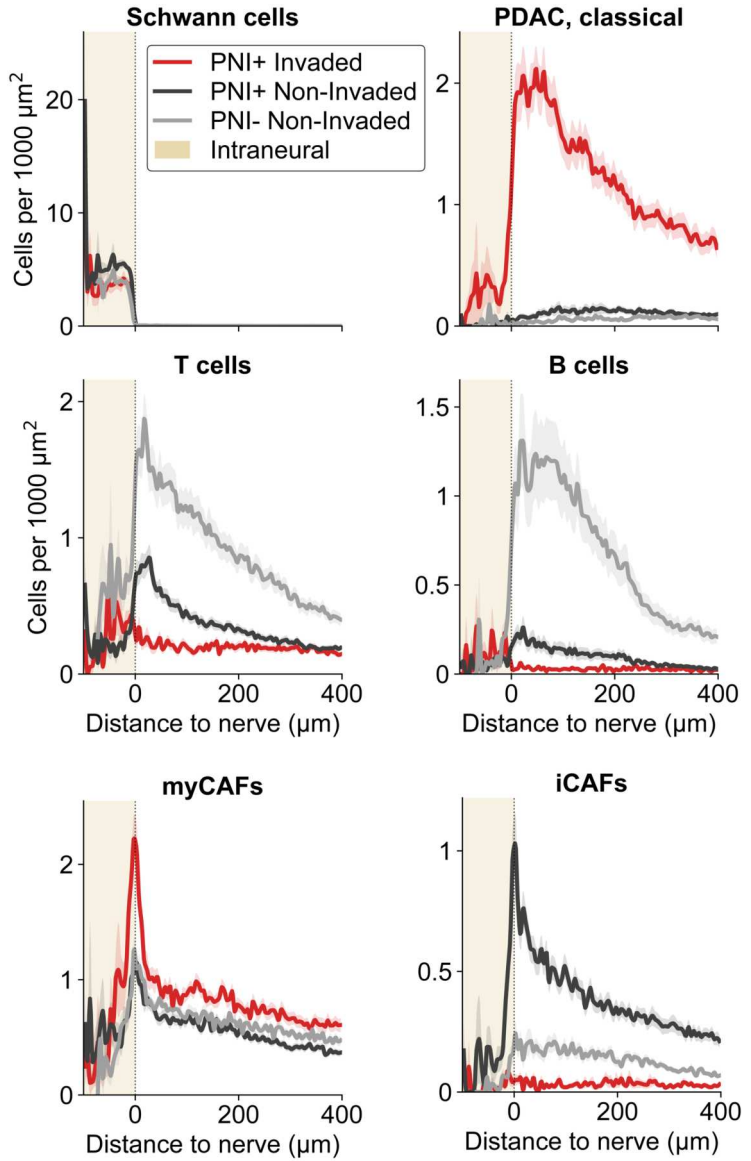


Does invasion of cancer cells alter the perineural niche?

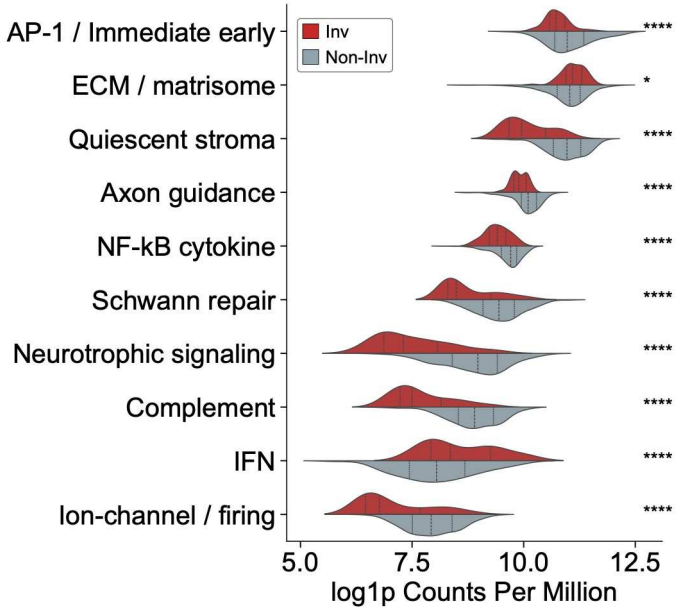


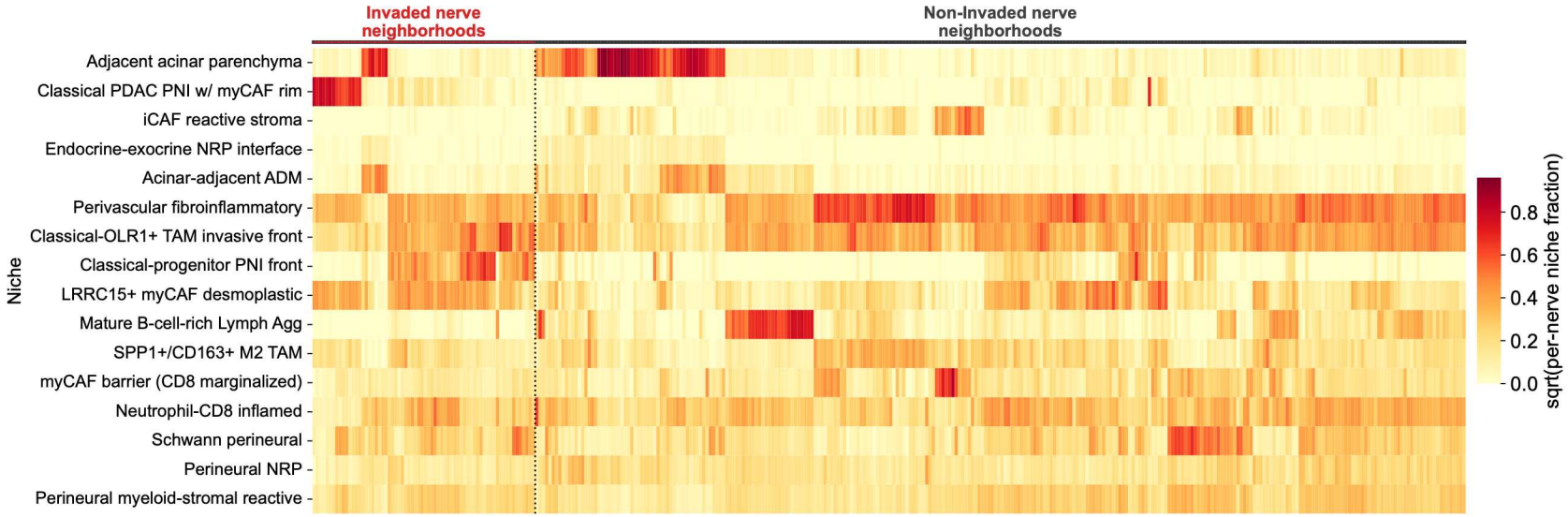
Nerve invasion is associated with reduced lymphocyte infiltration





Nerve invasion engages neurotrophic, matrix-remodeling, and lipid-handling programs





Summary 3: Invasion status stratifies the composition of PDAC perineural niche

- Perineural invasion in PDAC is associated with coordinated remodeling across malignant, stromal, immune, and neural compartments.

- Spatial & snRNA-seq atlases of PDAC nominate candidate perineural interactions and molecular programs for experimental validation and therapeutic development.

- Limitations
  - Small cohort size, particularly the PNI-low group, constrains patient-level generalizability.
  - Spatial associations remain descriptive and require orthogonal and functional validation.

Collaborative contributions

- Nicheverse is currently being applied to study
  - Dormant cancer cells in breast cancer (w/ Ines, Massagué lab)
  - Fibrotic niches in lung adenocarcinoma (w/ Elena, Massagué lab)
  - Immune triads in clear cell renal cell carcinoma (w/ Hui, Hakimi lab)

- Chan J, ...**Yarlagadda D**,...Rudin C. **Signatures of plasticity, metastasis, and immunosuppression in an atlas of human small cell lung cancer.** *Cancer Cell.* 2021;39(11):1479–1496.e18. doi: 10.1016/ j.ccell.2021.09.008.

- Lee J, …**Yarlagadda D**,… Massagué J. **TGF-β and RAS jointly unmask primed enhancers to drive metastasis.** *Cell*. 2024;187(22):6182–6199.e29. doi: 10.1016/j

Future directions

- Develop world models that predict **how cell states & tissue organization evolve over time & under perturbation**.

  ⟶ Resolve the metastatic cascade: from primary tumor evolution & dissemination to dormancy, metastatic outgrowth, treatment response, persistence, & relapse.

- With diverse multimodal, longitudinal & perturbational datasets, we can determine whether **recurrent niches** observed across biological contexts are governed by **shared mechanisms**.

- **Translate** insights into therapeutic vulnerabilities**, linking tumor evolution with AI-assisted therapeutic design,** including programmable gene and cell therapies, for rare cancers.